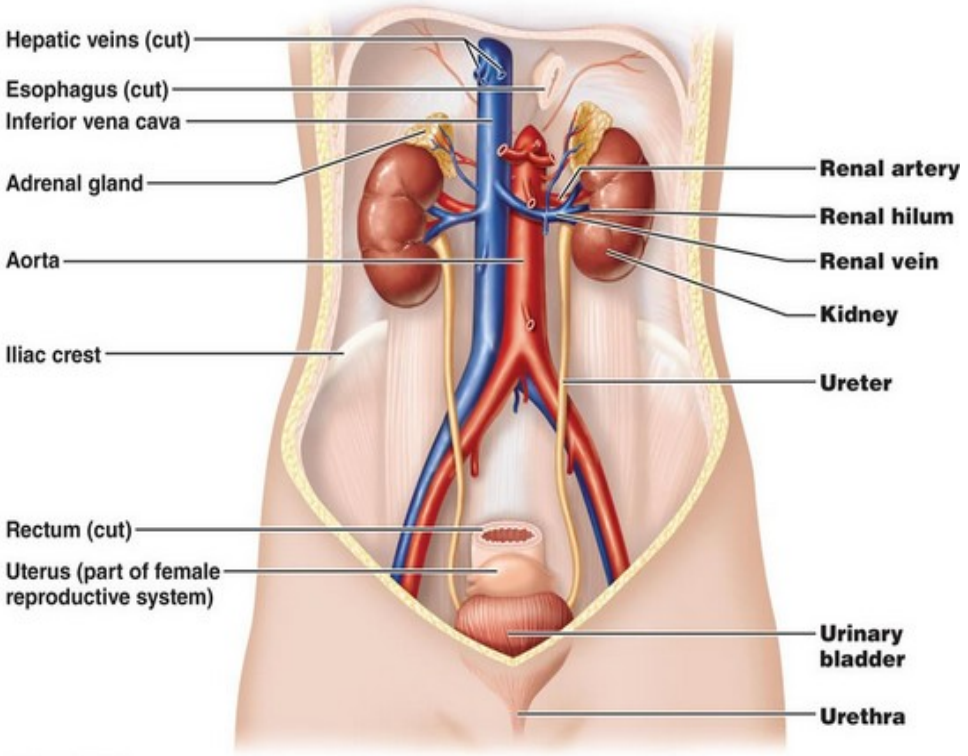
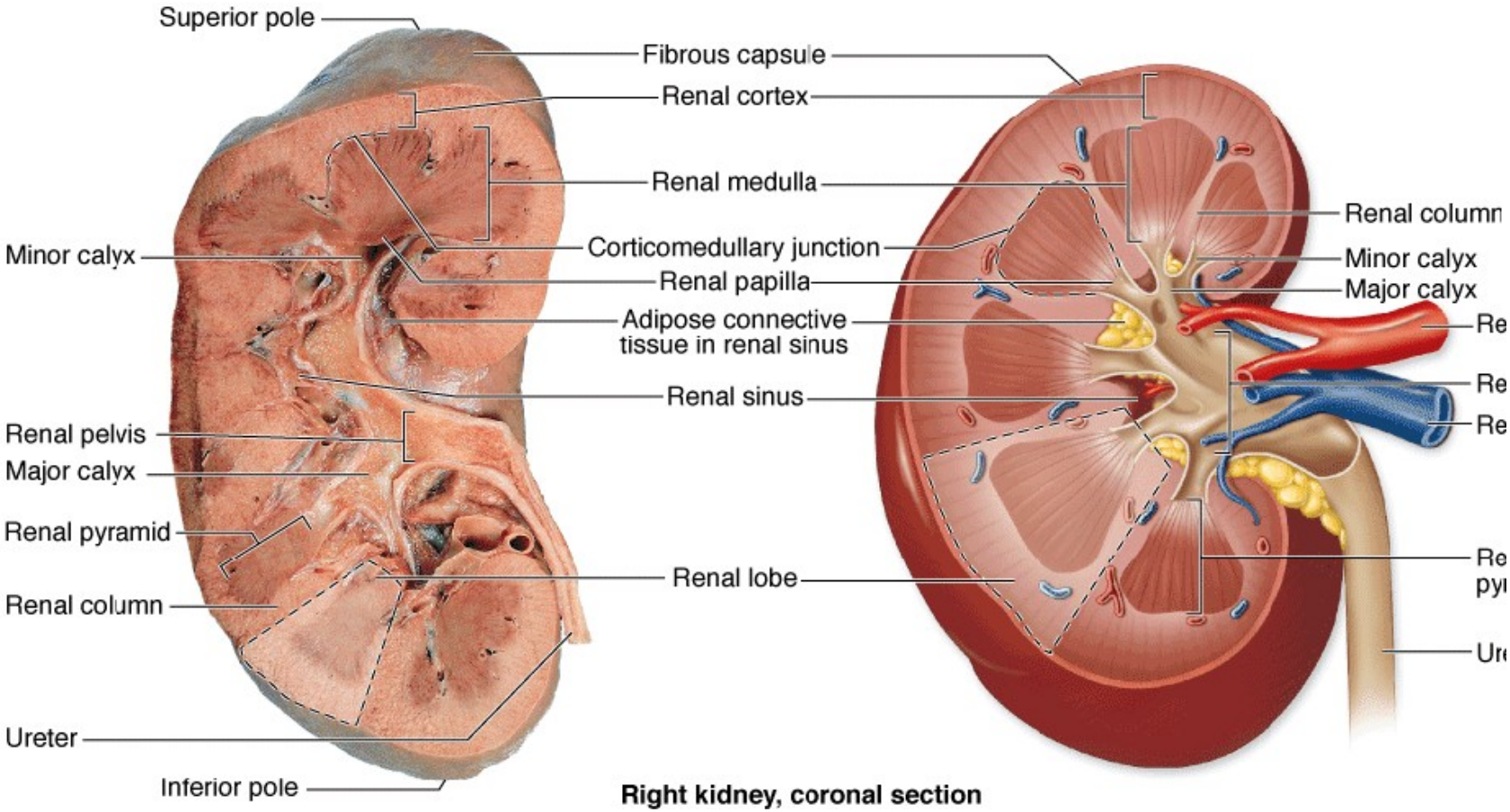

بسم الله الرحمن الرحيم

Introduction

- Each kidney is a large,bean-shaped organ located retroperitoneally on the posterior abdominal wall.
- It is covered by thin loose adhering capsule .
- On the medial side of the kidney is present a depression called hilum.
- The upper part of ureter is expanded to fill the hilum of kidney.This part is known as pelvis and is subdivided into large and small cups referred to as major calyces and minor calyces respectively.
- Kidneys remove the toxic watse products of metabolism but conserve water and essential electrolytes and metabolites(salts,glucose and proteins)





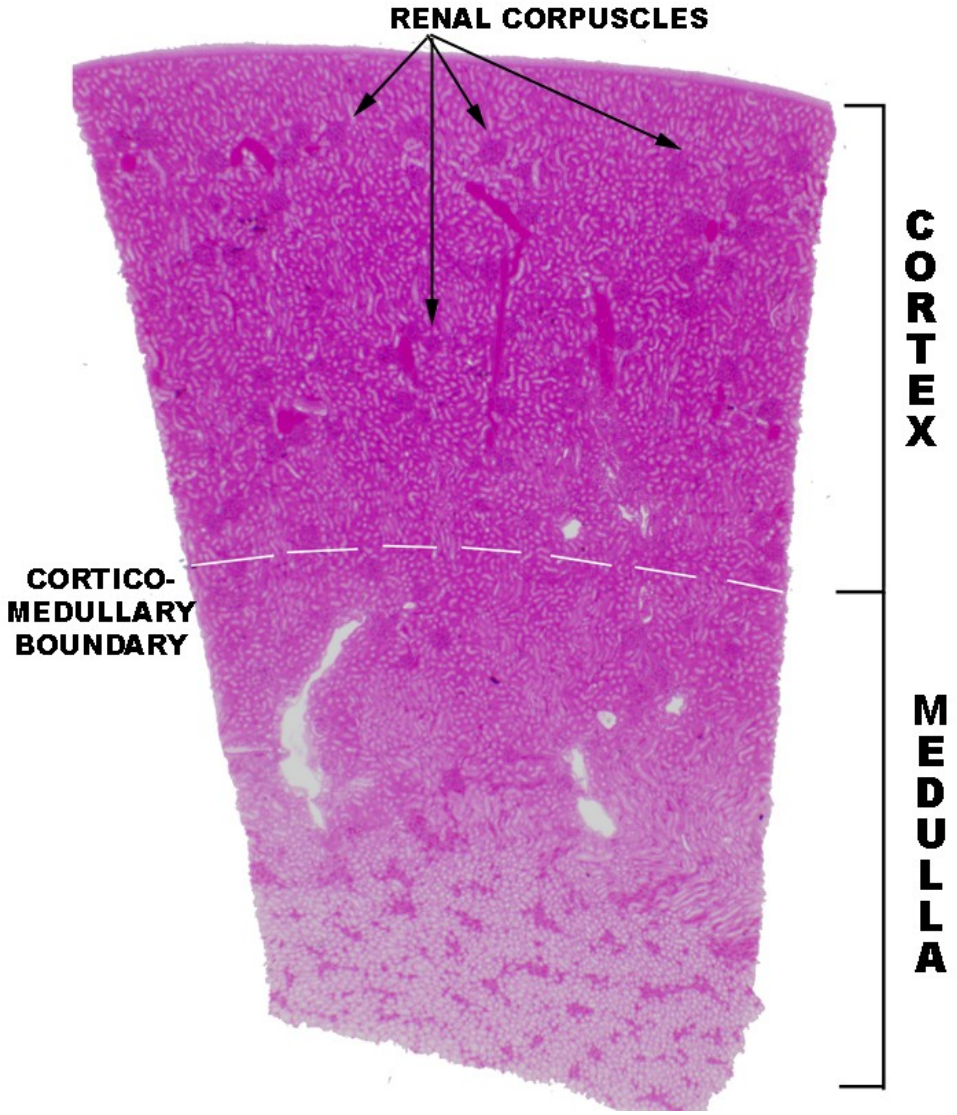
Right kidney, coronal section

Renal cortex

- The cortex is the outer region. The cortex has large numbers of **renal corpuscles**, discernible as round structures containing small blood vessels (90% blood passes through it).

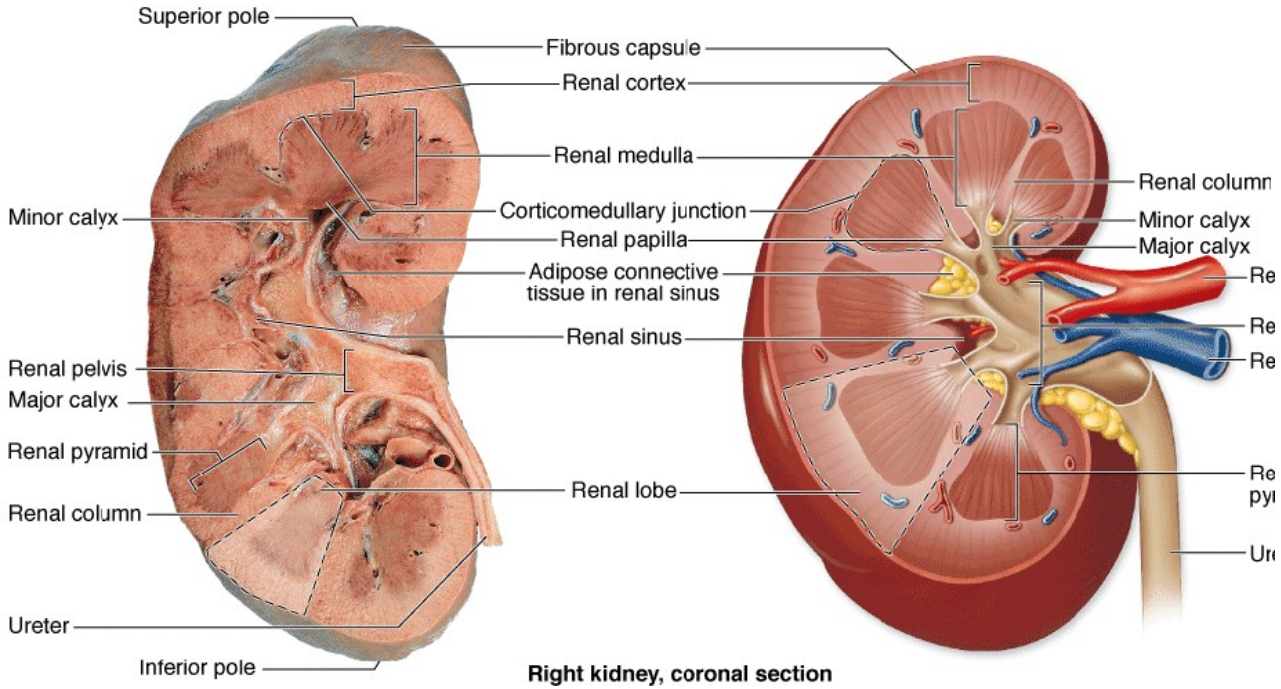
Renal Medulla

- The medulla gives **striated appearance** & its deep to the cortex, it has no renal corpuscles, only tubules are present. (10% blood passes through it).
- **Renal Pyramids.**
- **Renal columns (of Bertin).**
- The medulla has no renal corpuscles, but the cortex does have tubules.

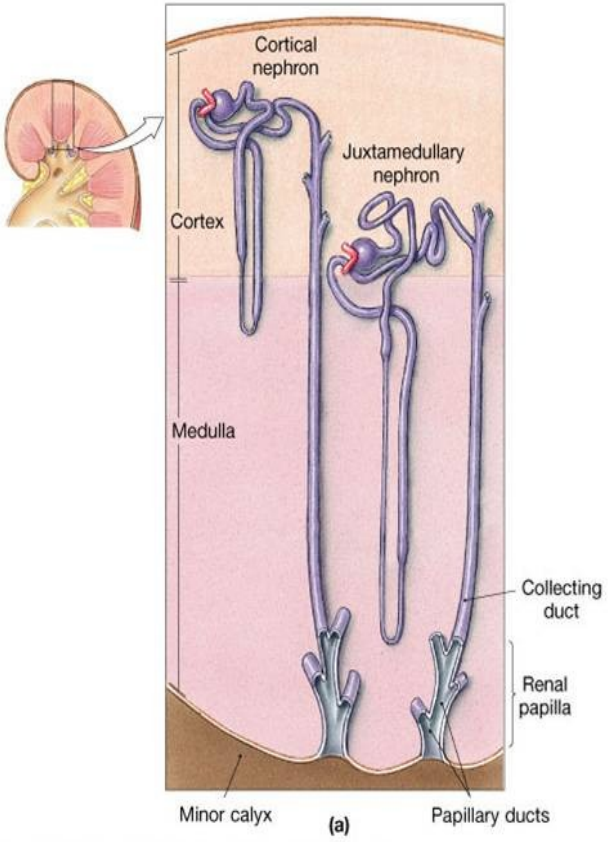


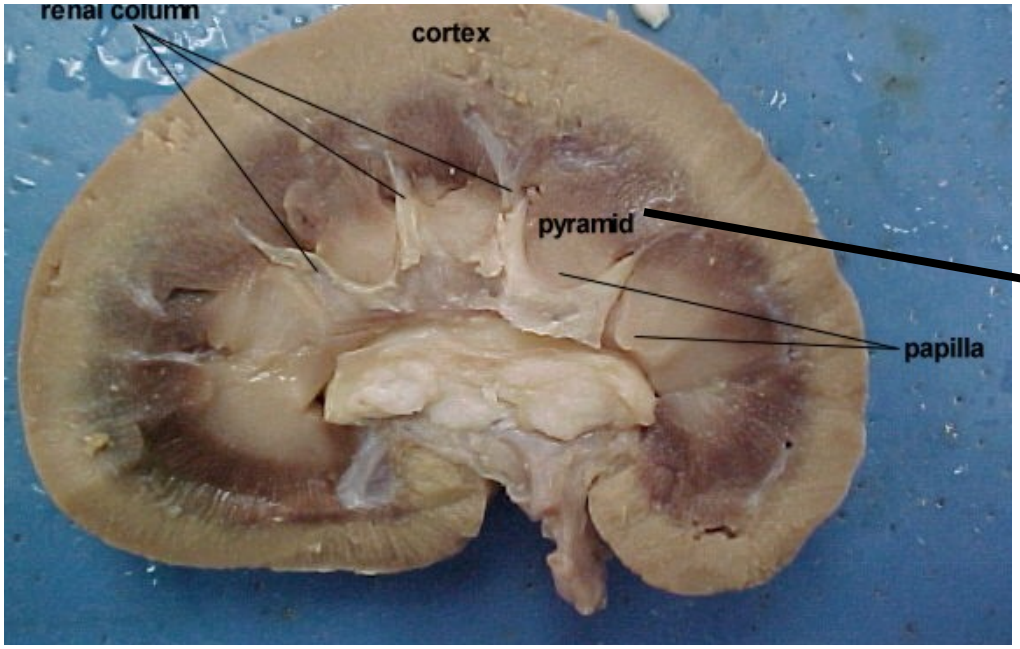
The cortex consists of *renal corpuscles* along with the *convoluted* and *straight tubules* of the *nephron*, the *collecting tubules*, *collecting ducts*, and an extensive vascular

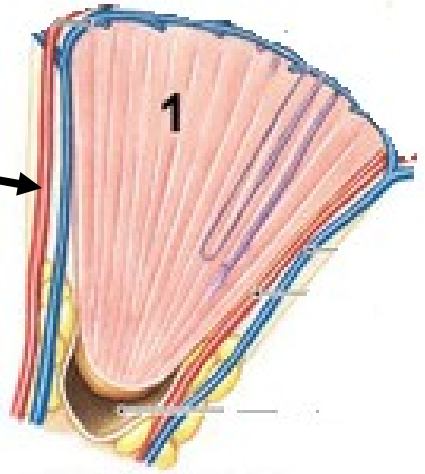
The renal corpuscles are spherical structures, barely visible with the naked eye. They constitute the beginning segment of the nephron and contain a unique capillary network called a *glomerulus*.



Right kidney, coronal section

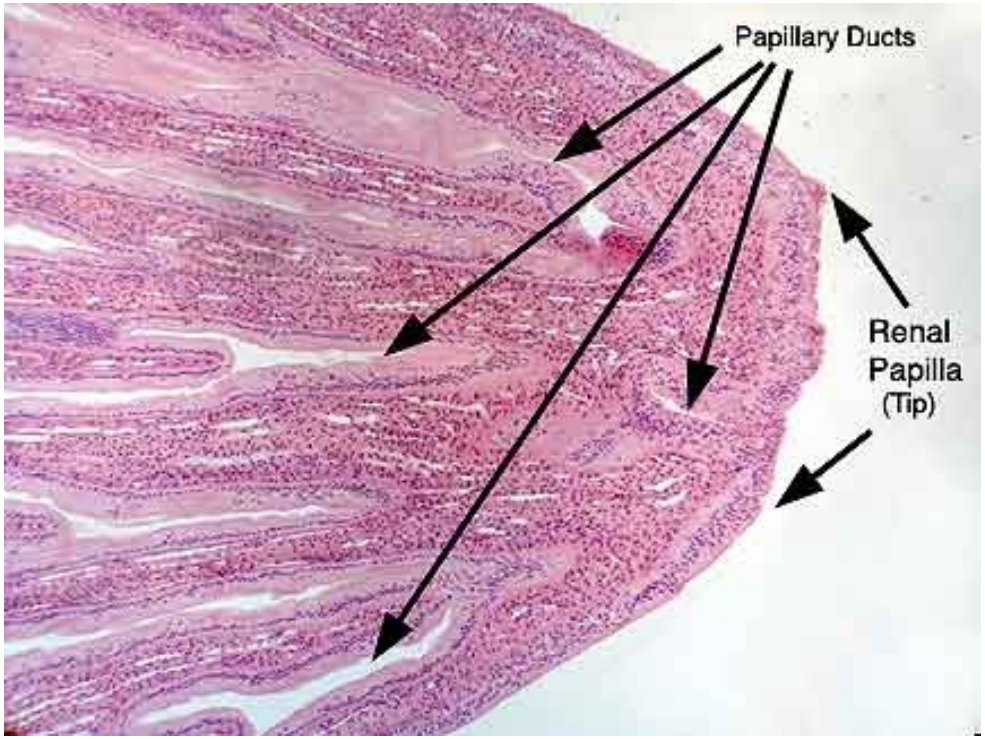


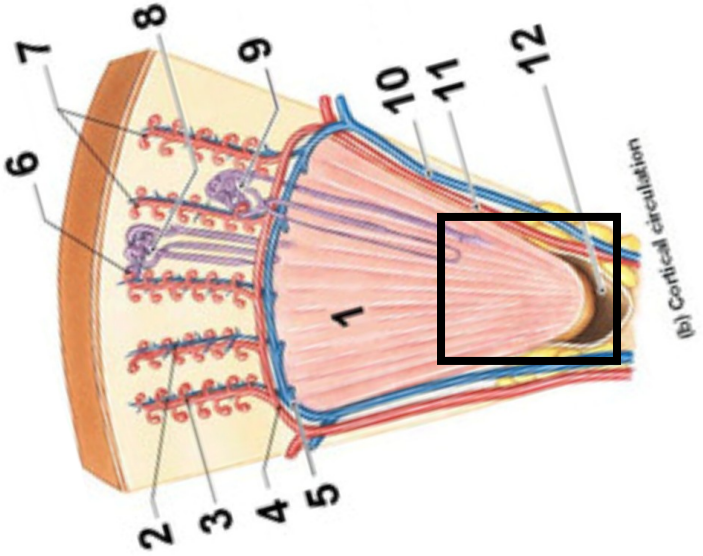




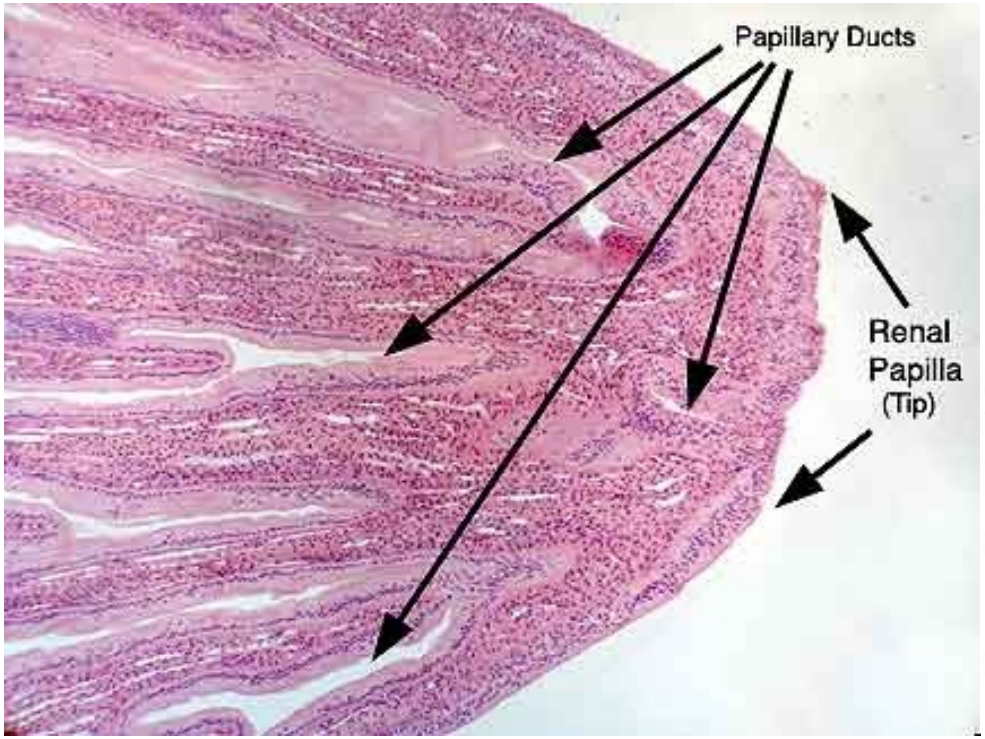
The tubules in the medulla, because of their arrangement and differences in length, collectively form a number of conical structures called *pyramids*. Usually 8 to 12 but as many as 18 pyramids may be present in the human kidney. The bases of the pyramids face the cortex, and the apices face the renal sinus.

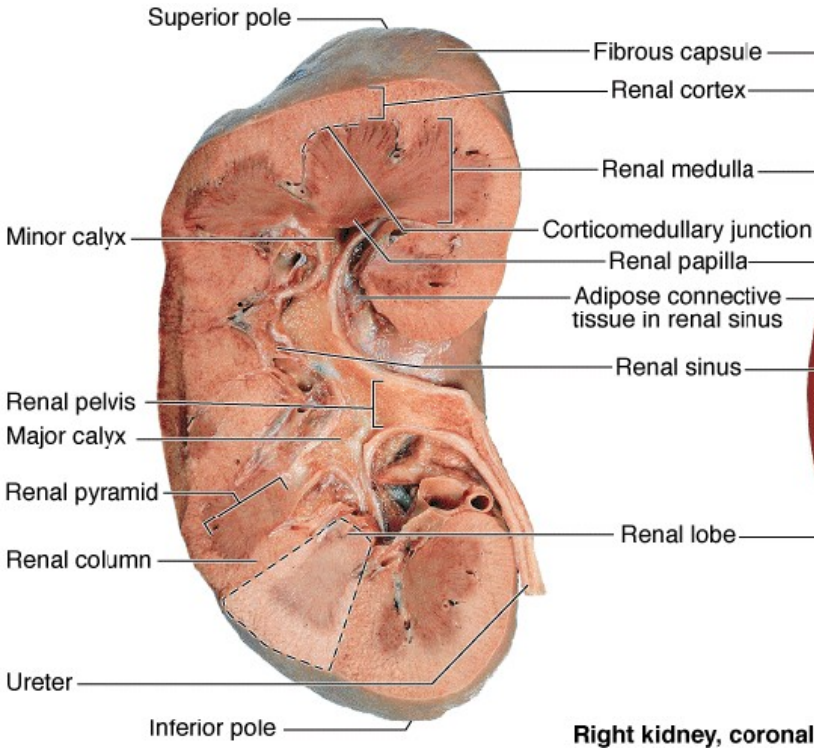
The **renal pyramids** are separated by cortical extensions called **renal columns (of Bertin)**.





The apical portion of each pyramid, which is known as the *papilla*, projects into a *minor calyx*, a cup-shaped structure that represents an extension of the renal pelvis. The tip of the papilla, also known as the *area cribrosa*, is perforated by the openings of the collecting ducts (Fig.





Right kidney, coronal

. The minor calyces are branches of the two or three *major calyces* that in turn are major divisions of the renal pelvis

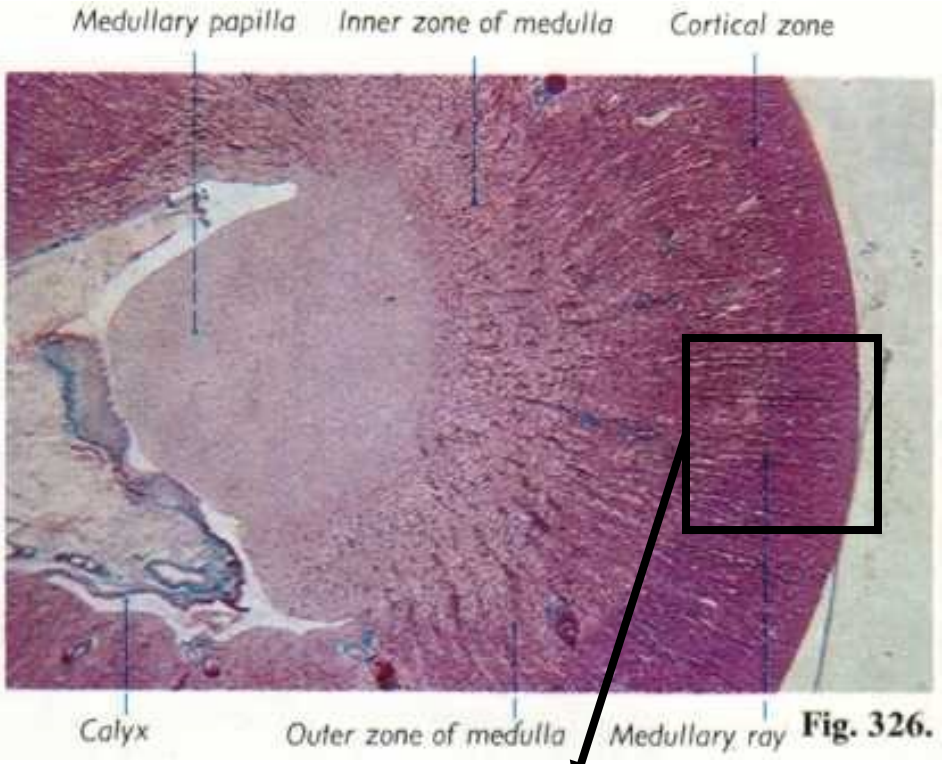
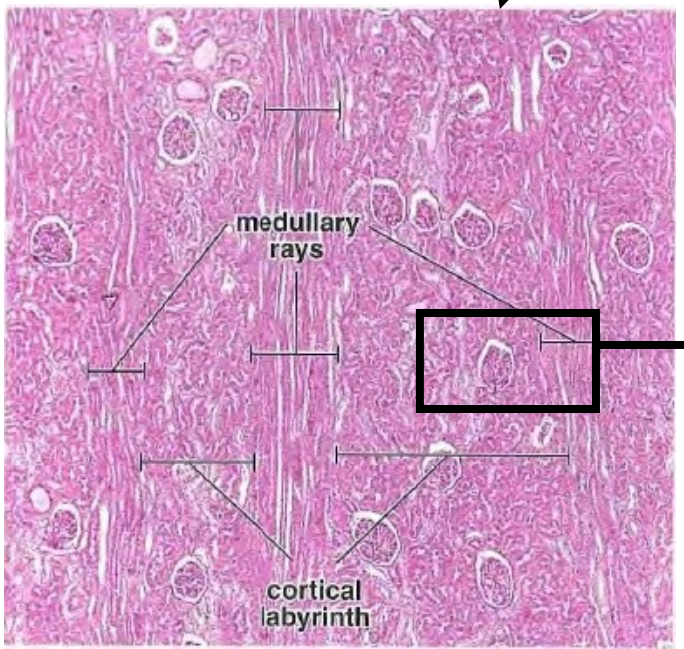
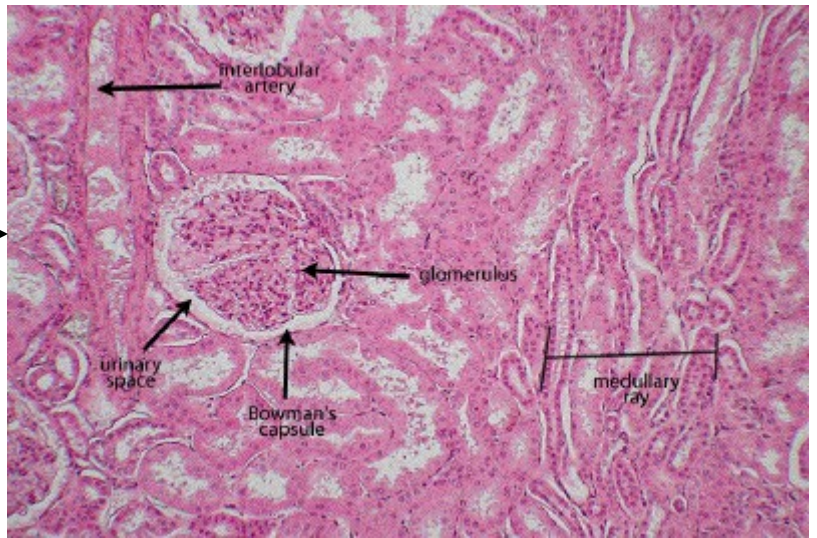
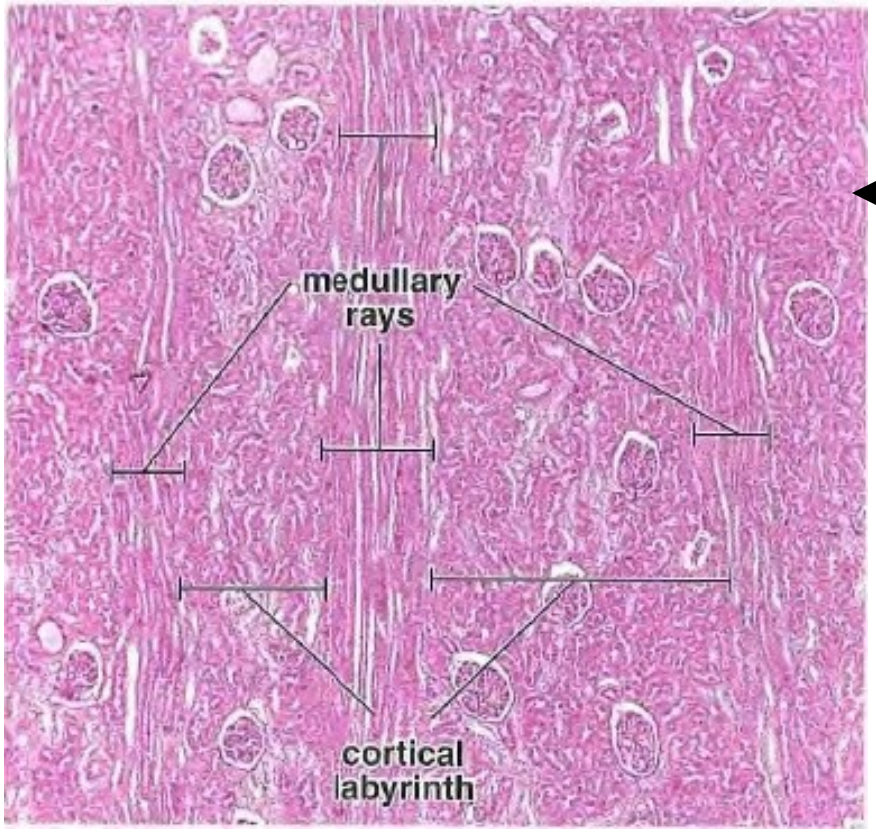

Medullary papilla
Inner zone of medulla
Cortical zone
Calyx
Outer zone of medulla
Medullary ray

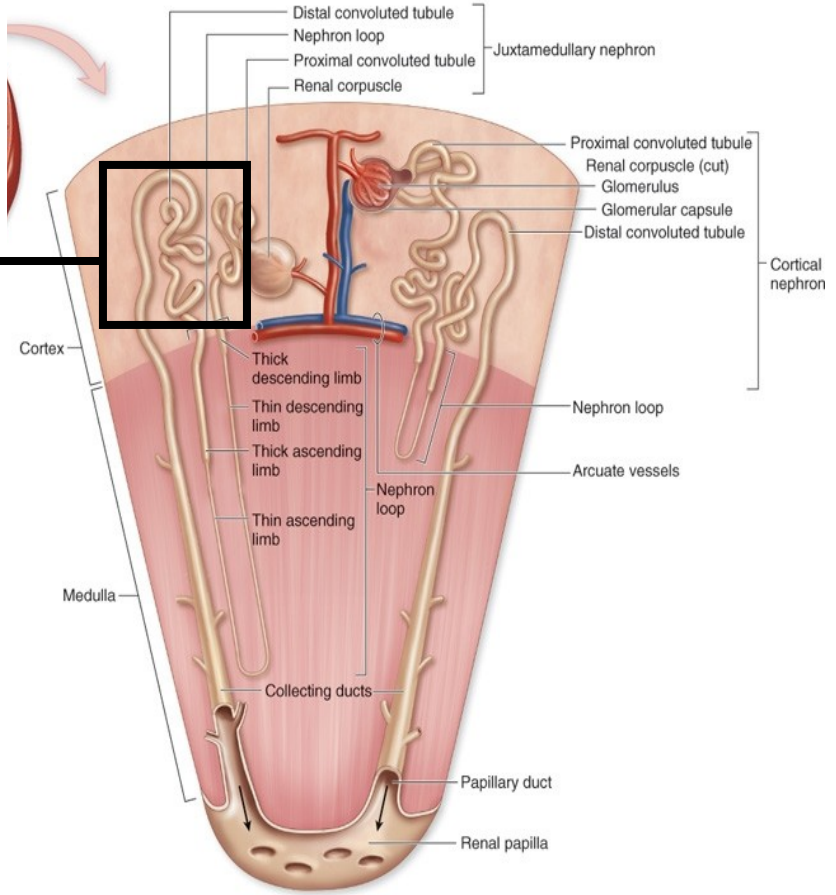
Fig. 326.


medullary rays
cortical labyrinth

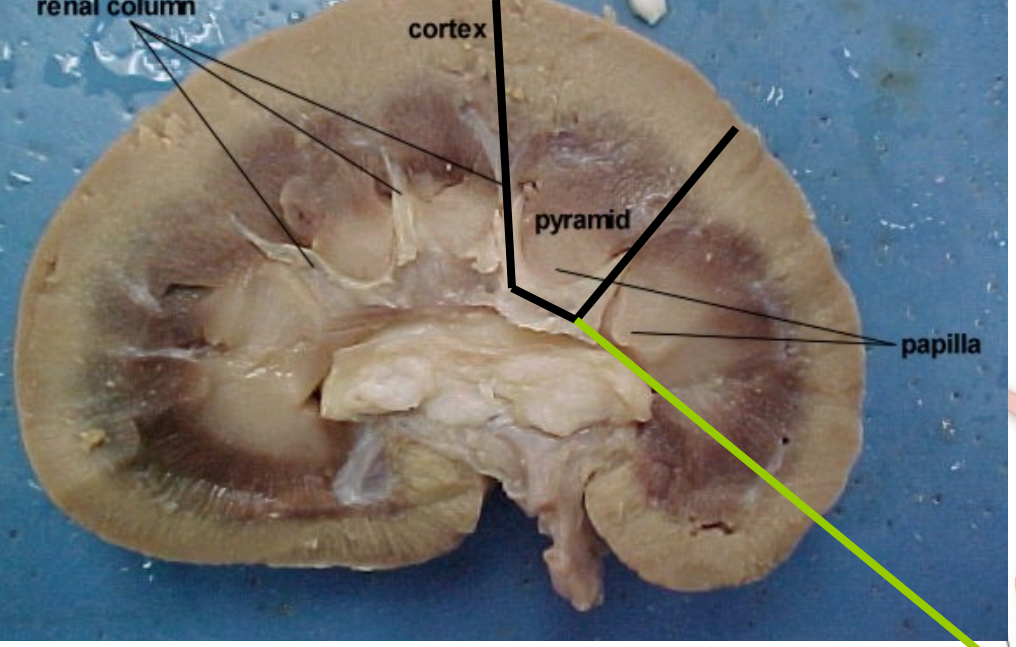

interlobular artery
glomerulus
urinary space
Bowman's capsule
medullary ray

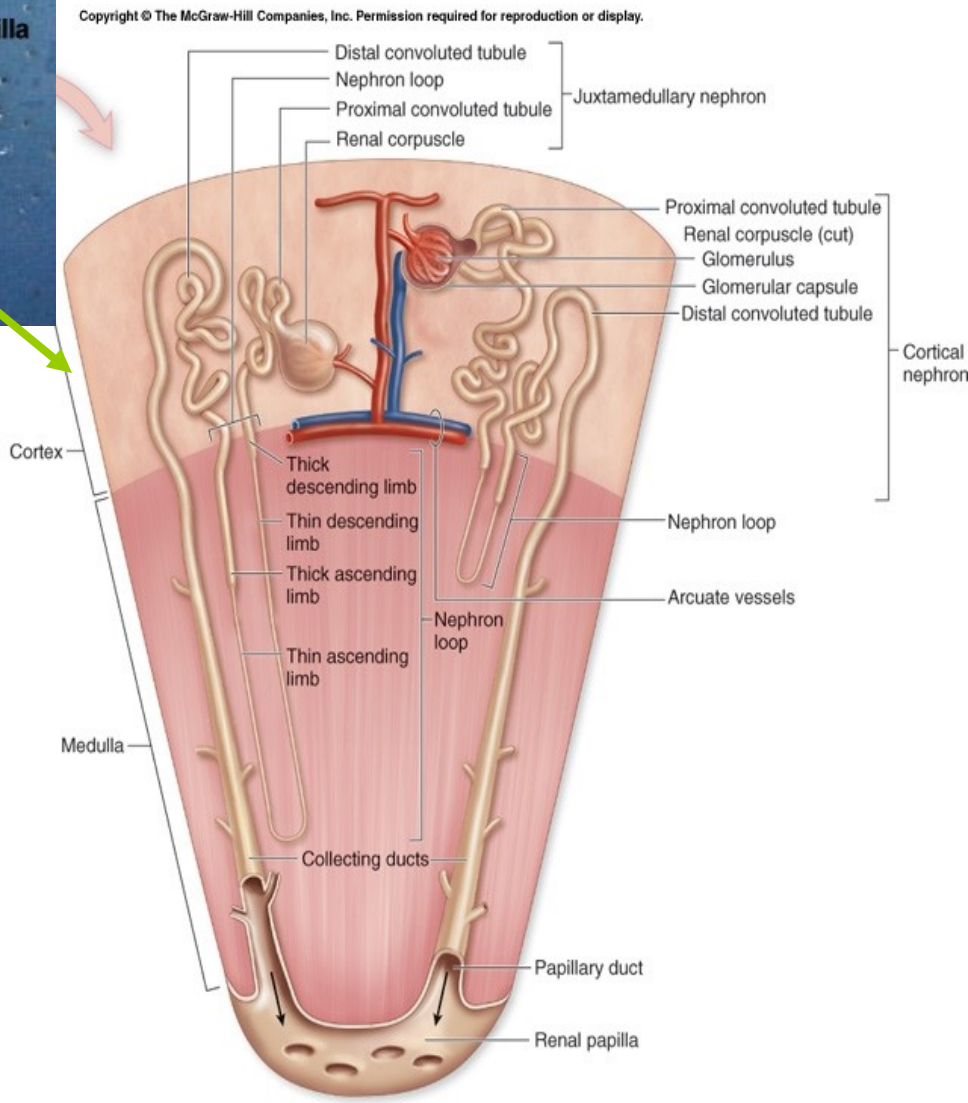


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Each medullary ray contains *straight tubules* of the nephrons and *collecting ducts*. The regions between medullary rays contain the renal corpuscles, the convoluted tubules of the nephrons, and the collecting tubules. These areas are referred to as *cortical labyrinths*. Each nephron and its *collecting tubule* (which connects to a *collecting duct* in the medullary ray) form the *uriniferous tubule*.

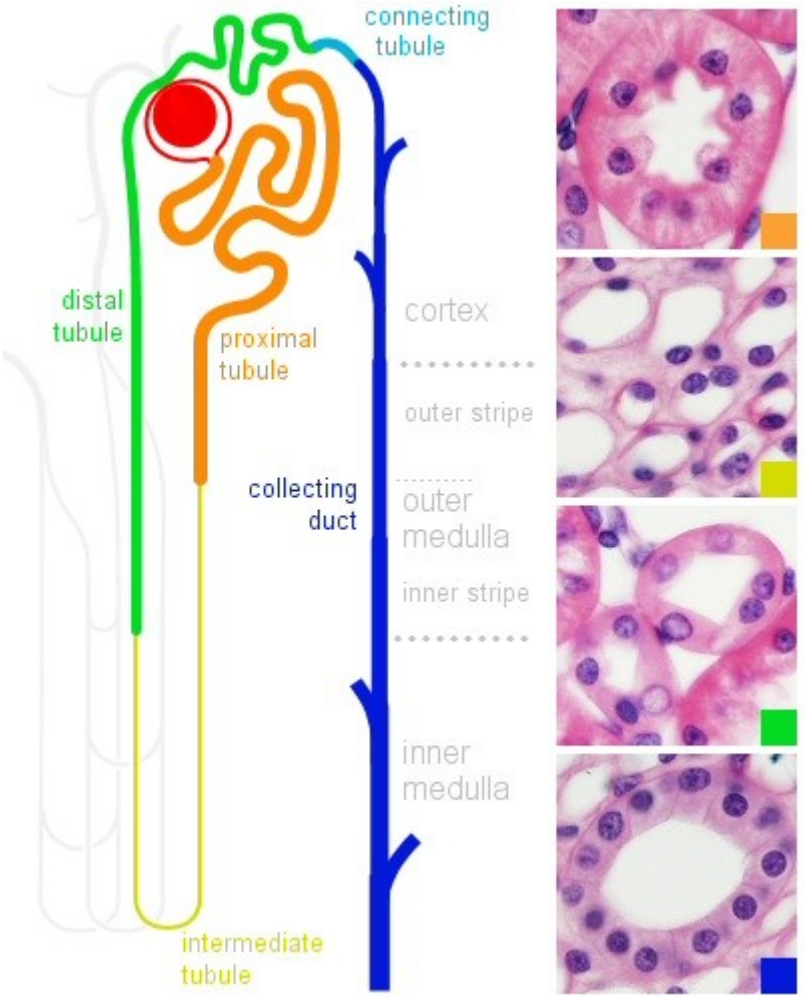




Each medullary pyramid plus the cortical tissue at its base and along its sides constitutes a **renal lobe**

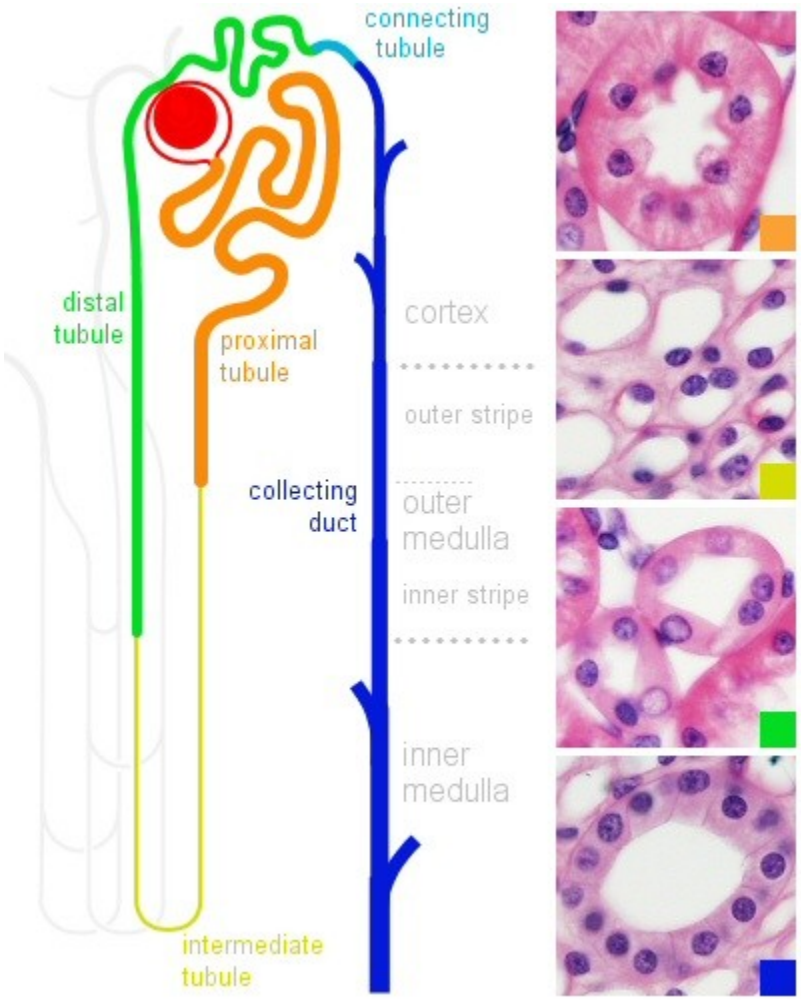
Urineferous tubule

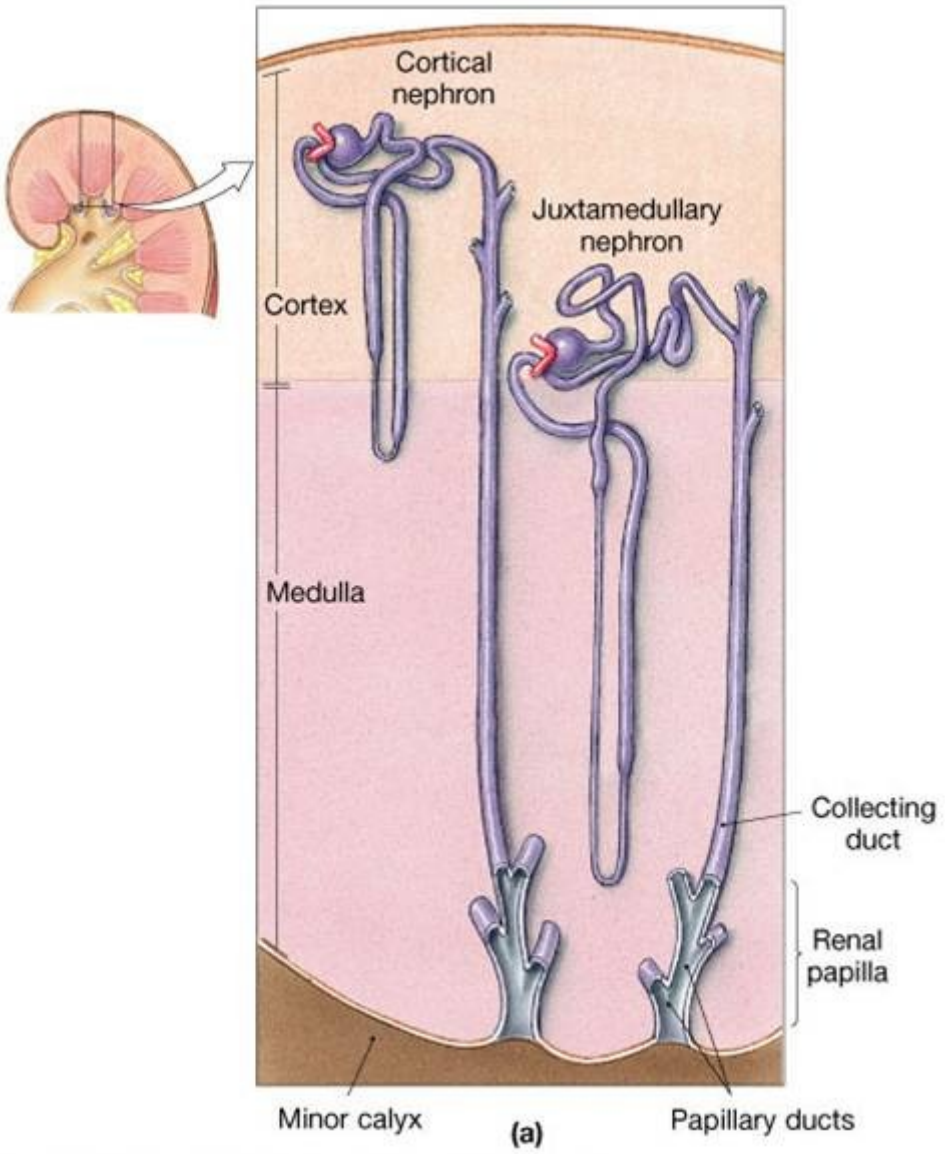
- A urineferous tubules forms structural and functional unit of kidney and consists of two parts:
- **Nephron:** is the longer part and is responsible for the production of urine.
- **Collecting tubule:**conveys urine from nephron to a collecting duct lying in a medullary ray.



Nephron

- Each kidney contains approximately 1.5million nephrons.
- A nephron is a long,epithelium lines tube which begins blindly and ends by joining a collecting tubule.
- Each nephron begins as a double walled cup shaped dilatation known as bowman's capsule**.**
- A tuft of capillaries called glomerulus,occupies the concavity of the capsule.
- The bowman's capsule and glomerulus collectively constitutes a renal corpuscle.
- Continuing from the bowman's capsule the remaining portions of nephron are:the proximal tubule,intermediate tubule and distal tubule





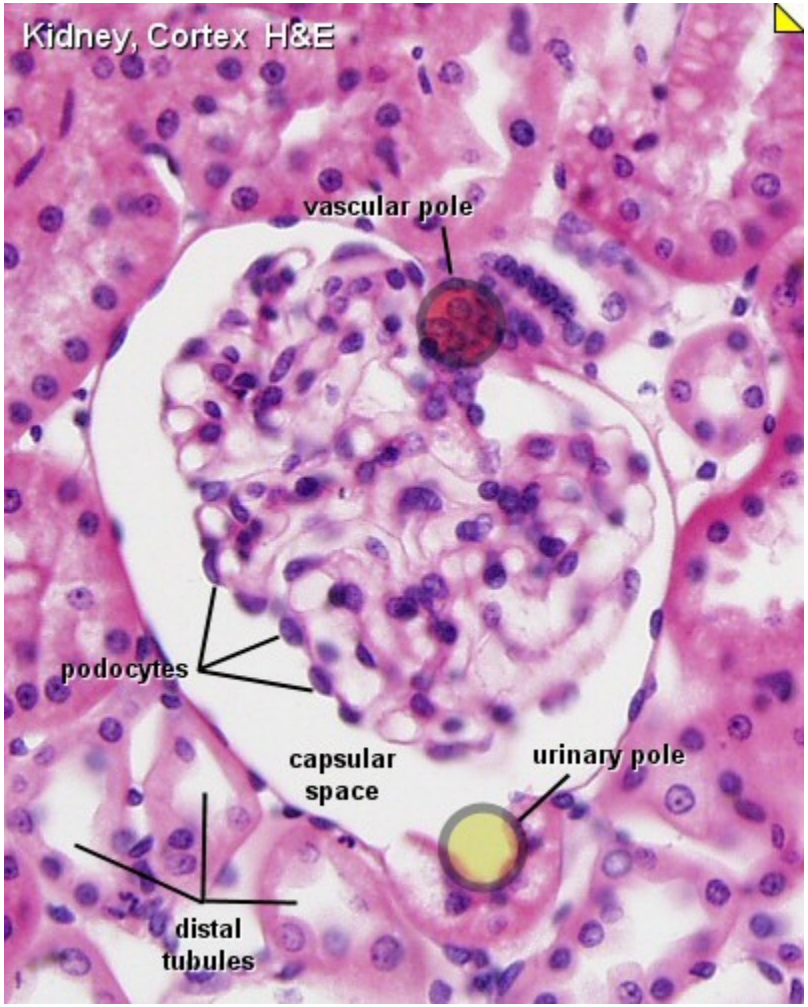
Types of Nephrons

Several types of nephrons are identified, based on the location of their renal corpuscles in the cortex (see Fig. 19.3):

- *Subcapsular* or *cortical nephrons* have their renal corpuscles located in the outer part of the cortex. They have short loops of Henle, extending only into the outer medulla. They are typical of the nephrons described above, wherein the hairpin turn occurs in the distal straight tubule.
- *Juxtamedullary nephrons* make up about one-eighth of the total nephron count. Their renal corpuscles occur in proximity to the base of a medullary pyramid. They have long loops of Henle and long ascending thin segments that extend well into the inner region of the pyramid. These structural features are essential to the urine-concentrating mechanism, which is described below.
- *Intermediate* or *midcortical nephrons* have their renal corpuscles in the midregion of the cortex. Their loops of Henle are of intermediate length.

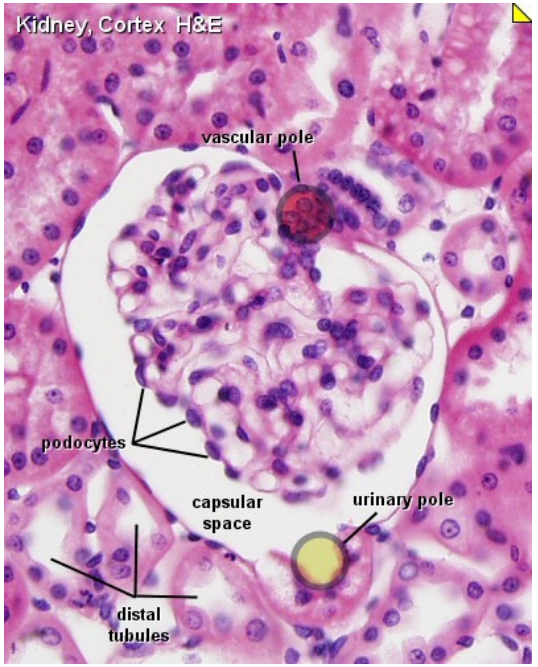
Renal corpuscle

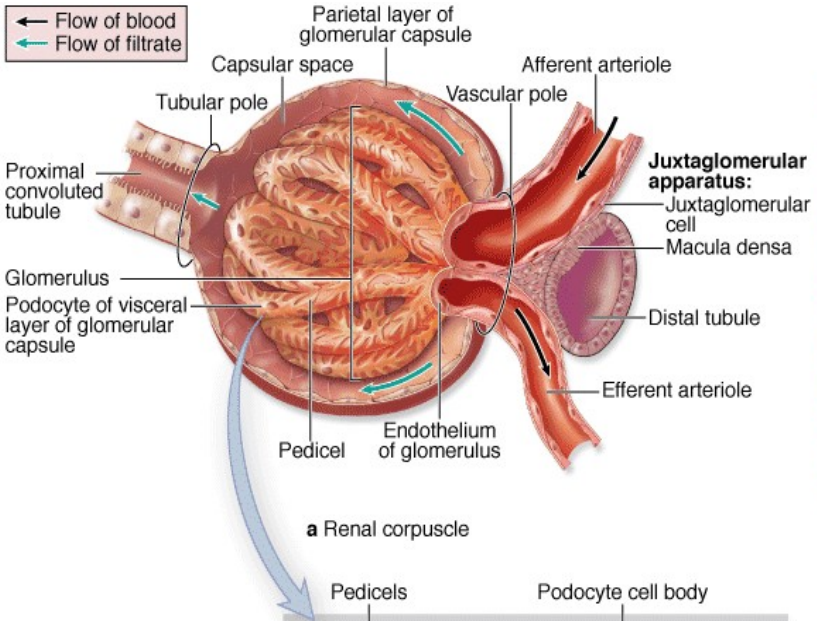
- **Renal Corpuscles & Blood Filtration**
- At the beginning of each nephron is a renal corpuscle (spherical bodies), about 200-250 micro m in diameter and containing a loose knot of capillaries, the glomerulus, surrounded by a double-walled cup shaped epithelial capsule called the **glomerular (Bowman's) capsule**

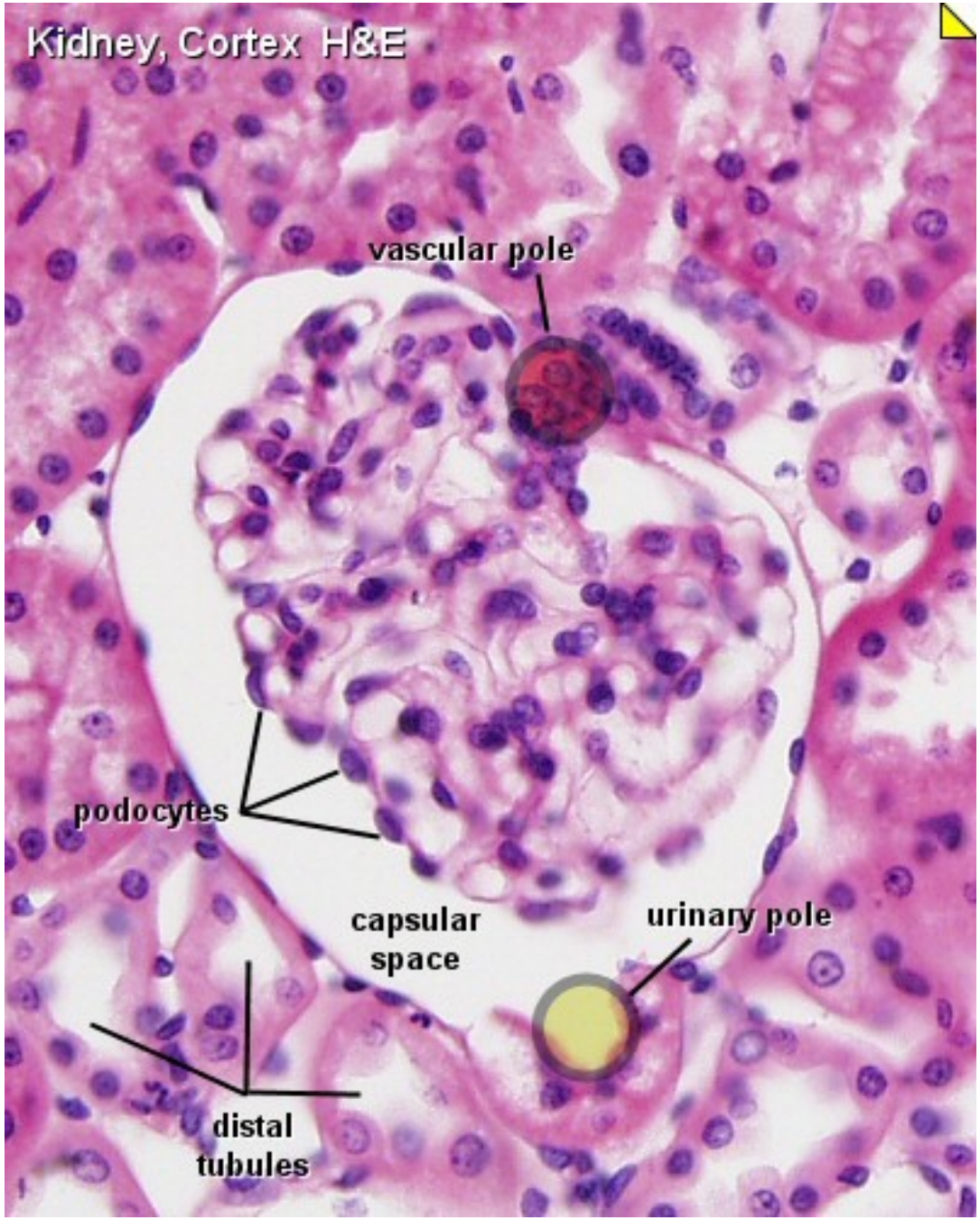


Glomerulus (Bowman's) capsule

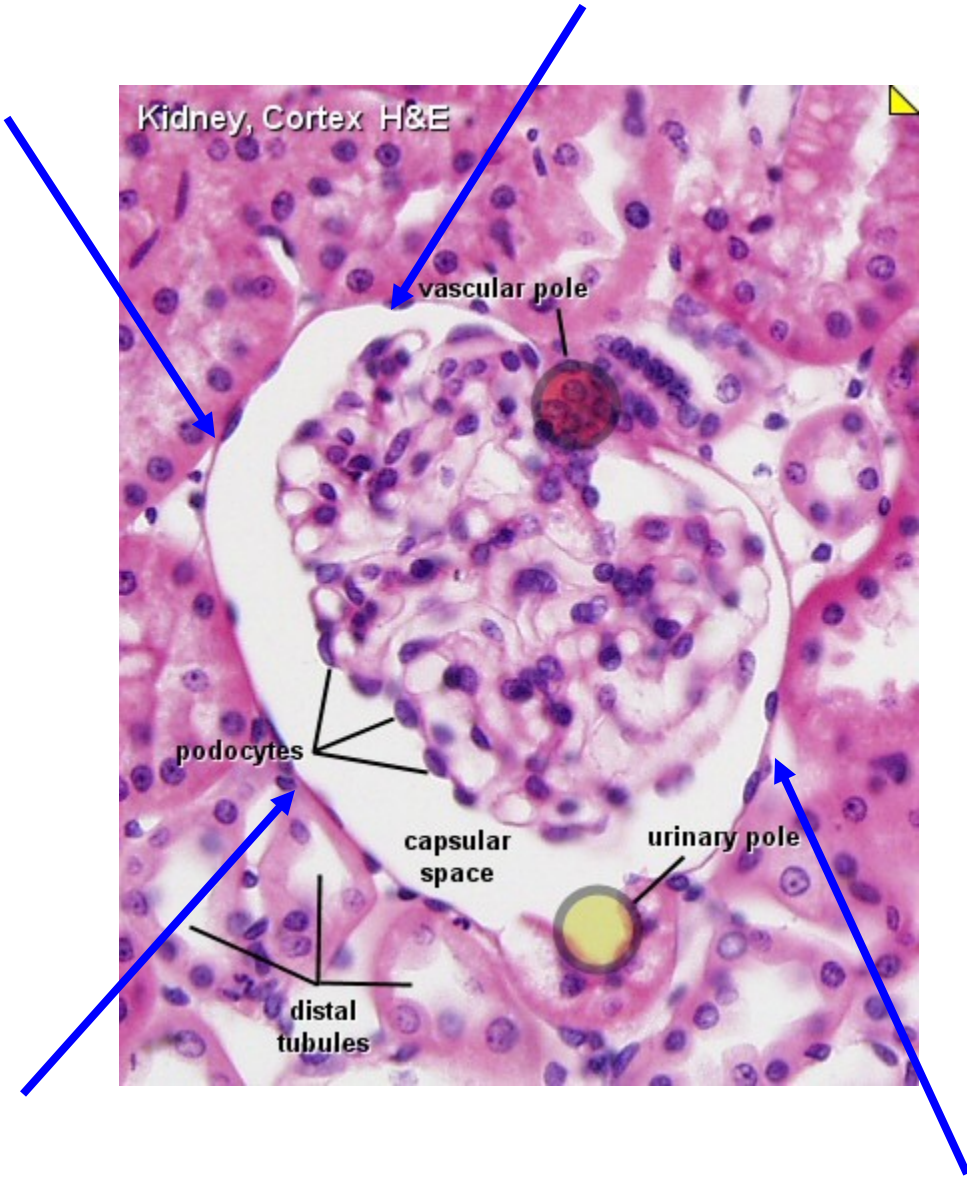
- The internal layer (**visceral layer**) of the capsule closely envelops the glomerular capillaries.
- The external **parietal layer** forms the outer surface of the capsule.
- Between the two capsular layers is the **urinary** or **capsular space,** which receives the fluid filtered through the capillary wall and the visceral layer.
- Each renal corpuscle has a **vascular pole,** where the afferent arteriole enters and the efferent arteriole leaves, and a urinary or **tubular pole,** where the proximal convoluted tubule begins.
- After entering the renal corpuscle, the afferent arteriole usually divides and subdivides into the two to five capillaries of the renal glomerulus.

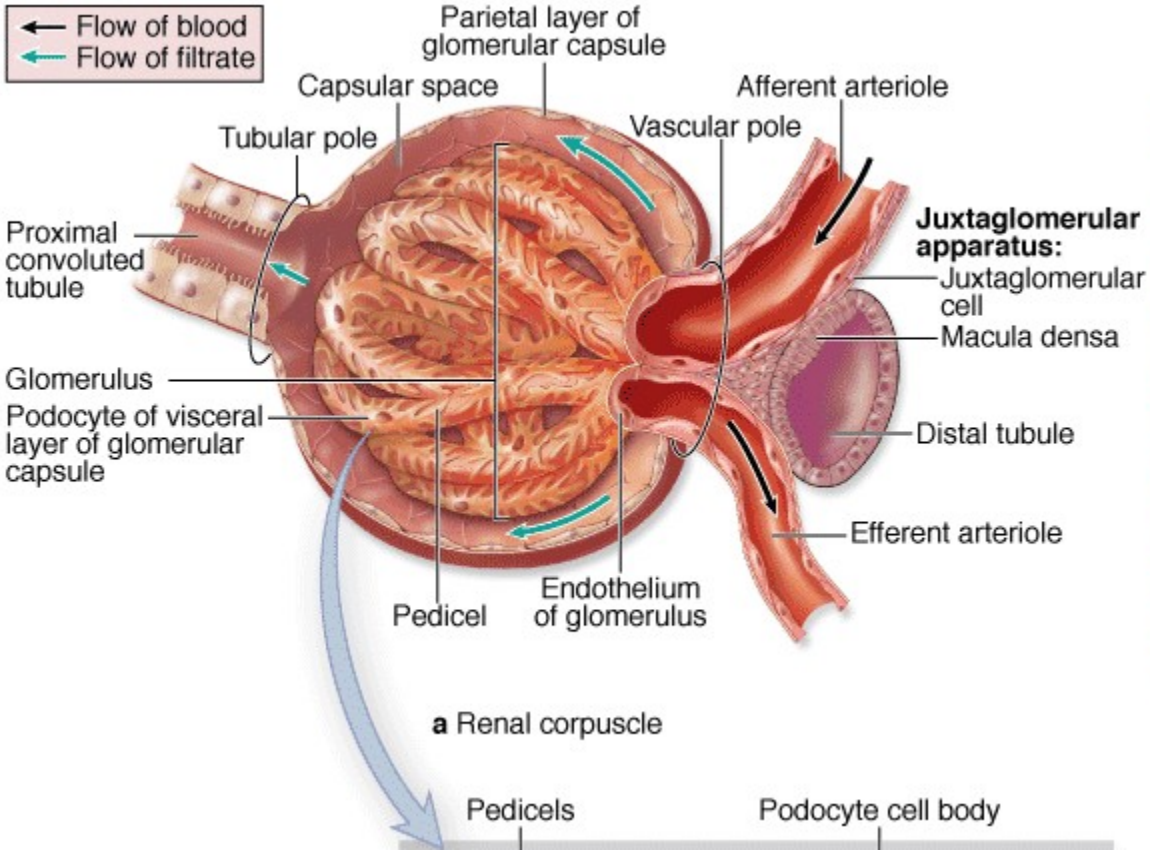




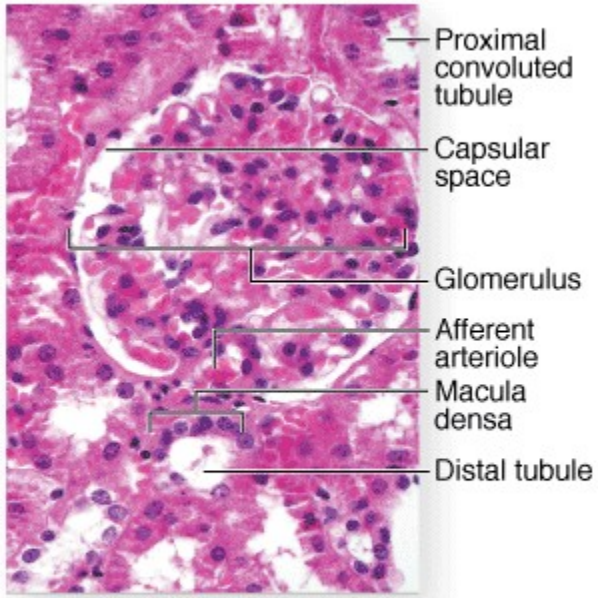

Kidney, Cortex H&E
vascular pole
podocytes
capsular space
urinary pole
distal tubules

- **Glomerulus (Bowman's) capsule**
- Cells of the outer or ***parietal layer*** of Bowman's capsule form a simple squamous epithelium which are supported externally by a basal lamina and a thin layer of reticular fibers. At the tubular pole, this epithelium changes to the **simple cuboidal epithelium** characteristic of the ***proximal convulated tubule***.



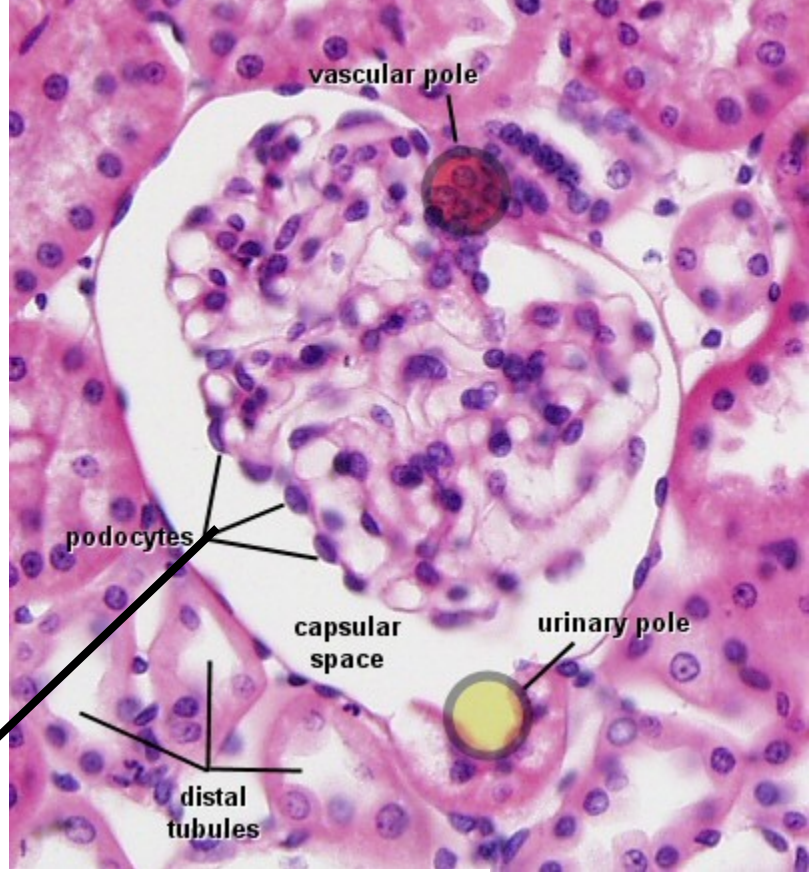


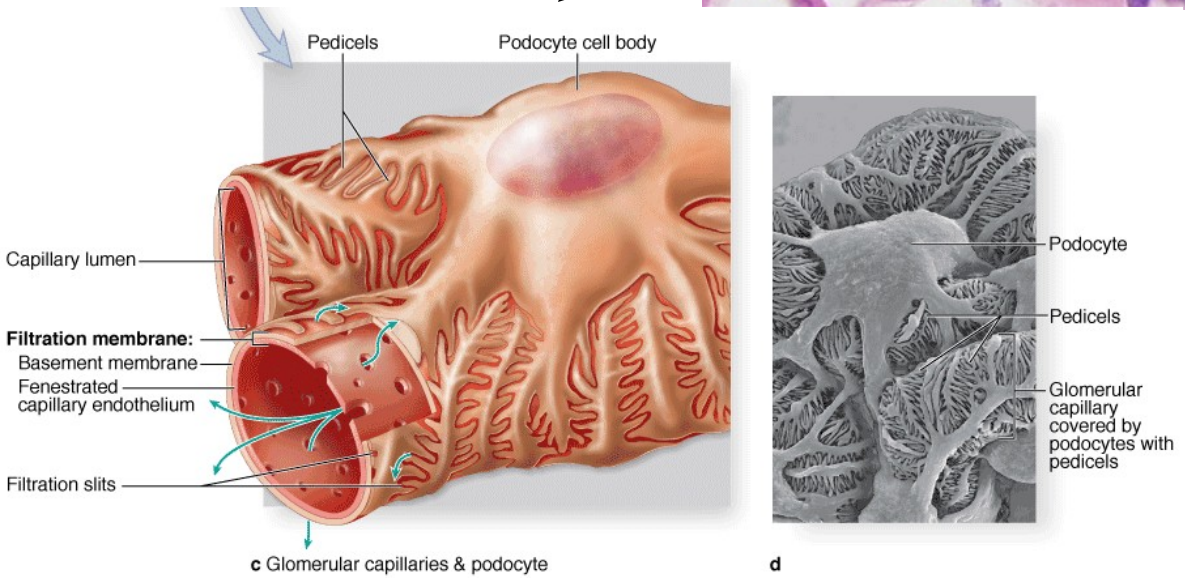
a Renal corpuscle



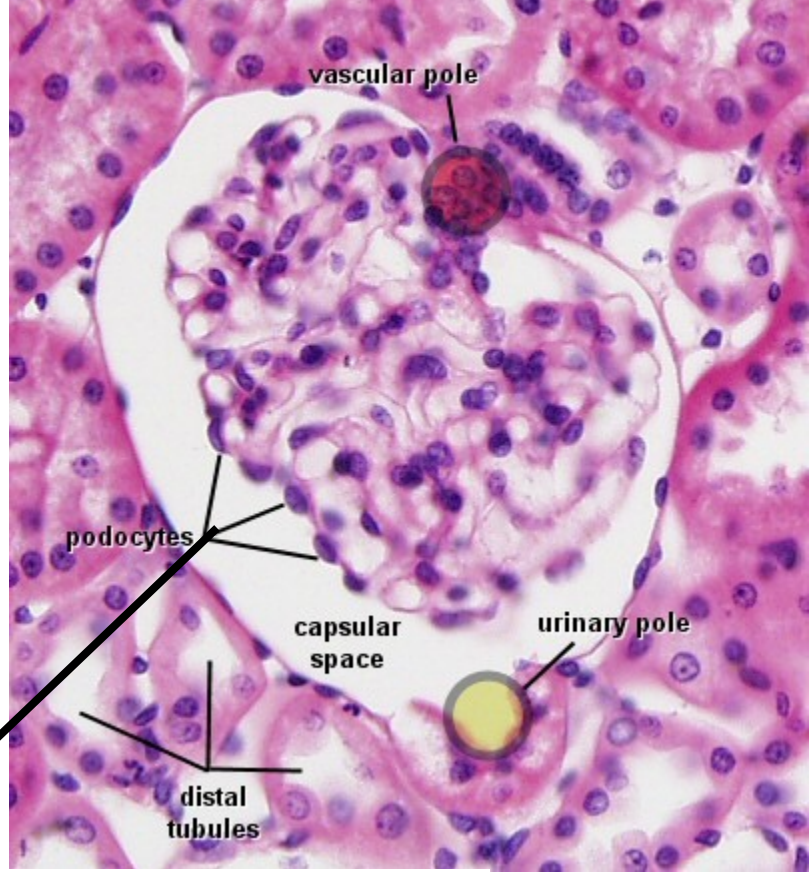
b Histology of renal corpuscle

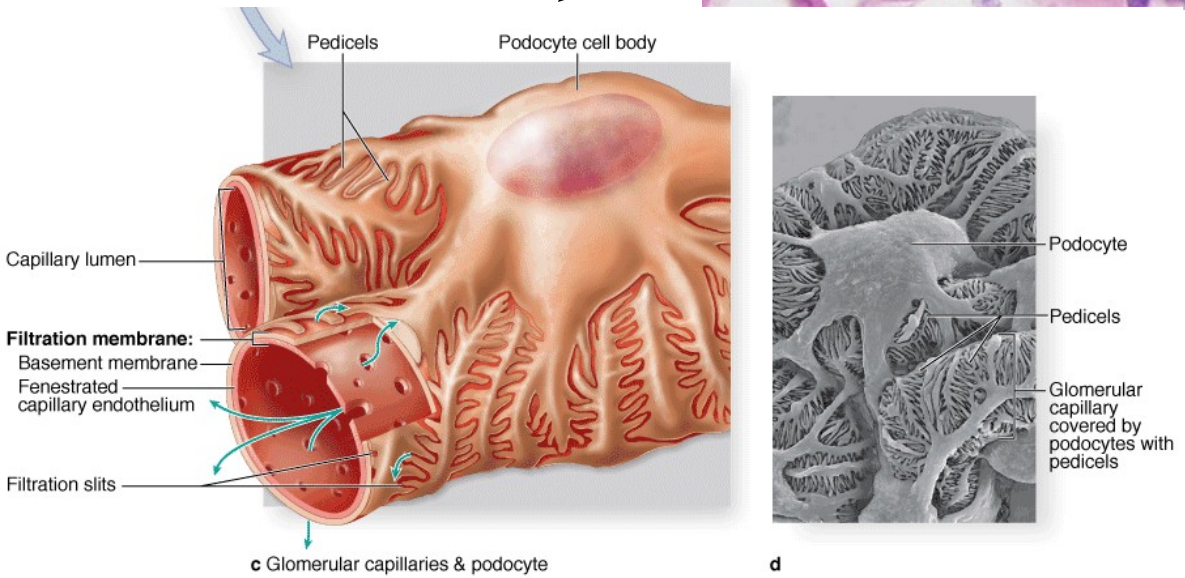
- **Glomerulus (Bowman's) capsule**
- Cells of the inner layer or **visceral layer** are called **<u>podocytes</u>**.
- Podocytes are large irregularly star shaped cells which share a common basal lamina with the endothelial cells of glomerular capillaries.
- Cell bodies of podocytes are located in urinary space. From the cell body arises several primary processes & each of these processes give rise to secondary processes called pedicles.
- Numerous microtubules & actin filaments are found in podocytes.

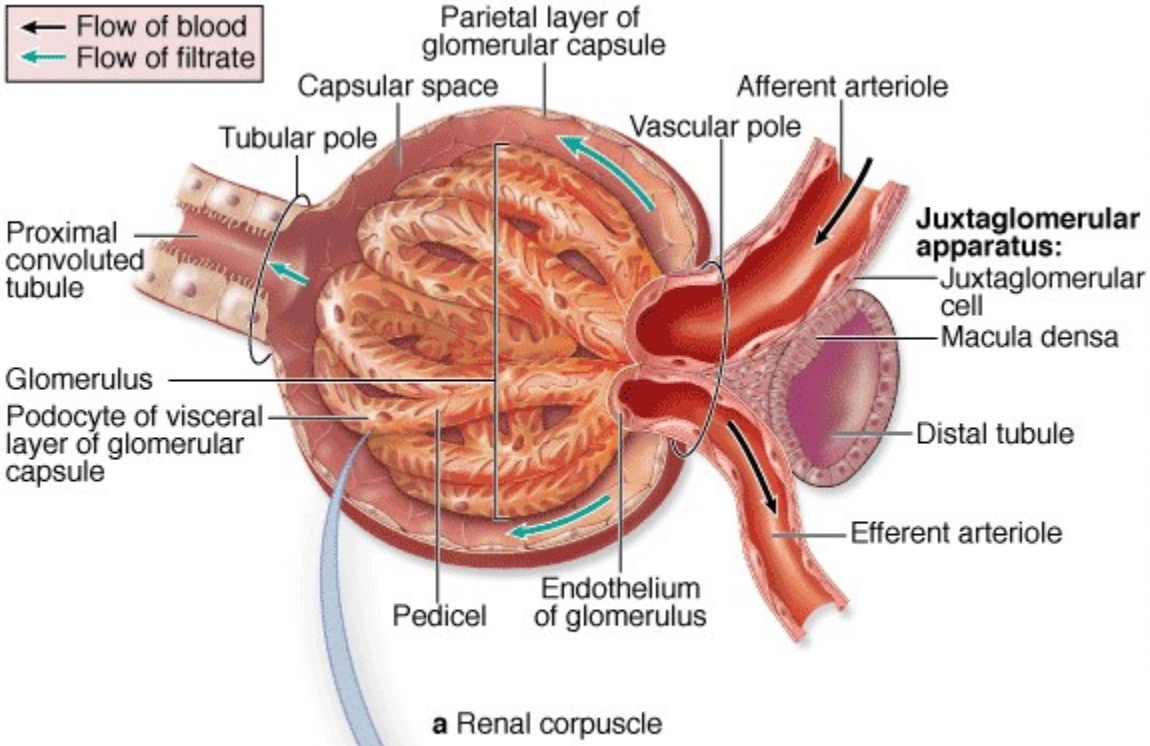




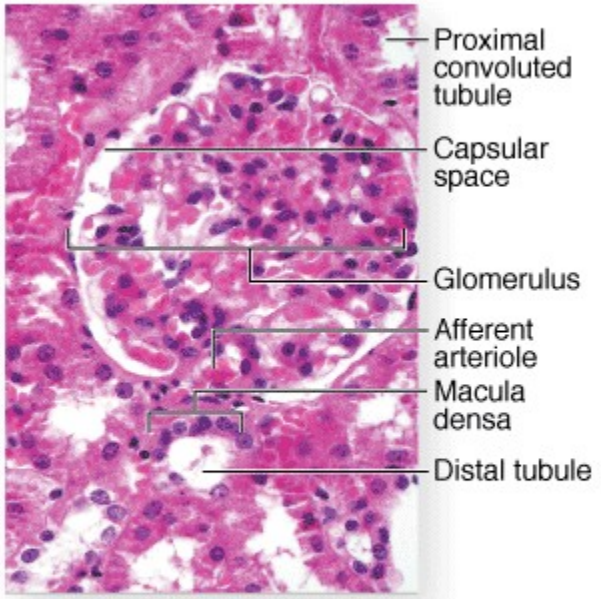
- **<u>Podocytes</u>**.
- Cytoplasm of pedicles form a fenestrated epithelium around the fenestrated capillaries of the glomerulus.
- The openings between the pedicles are called **filtration slits(25nm).** They are spanned by a thin membrane, the **filtration slit membrane or slit diaghragms(5nm thick having pores of 6nm in diameter)**



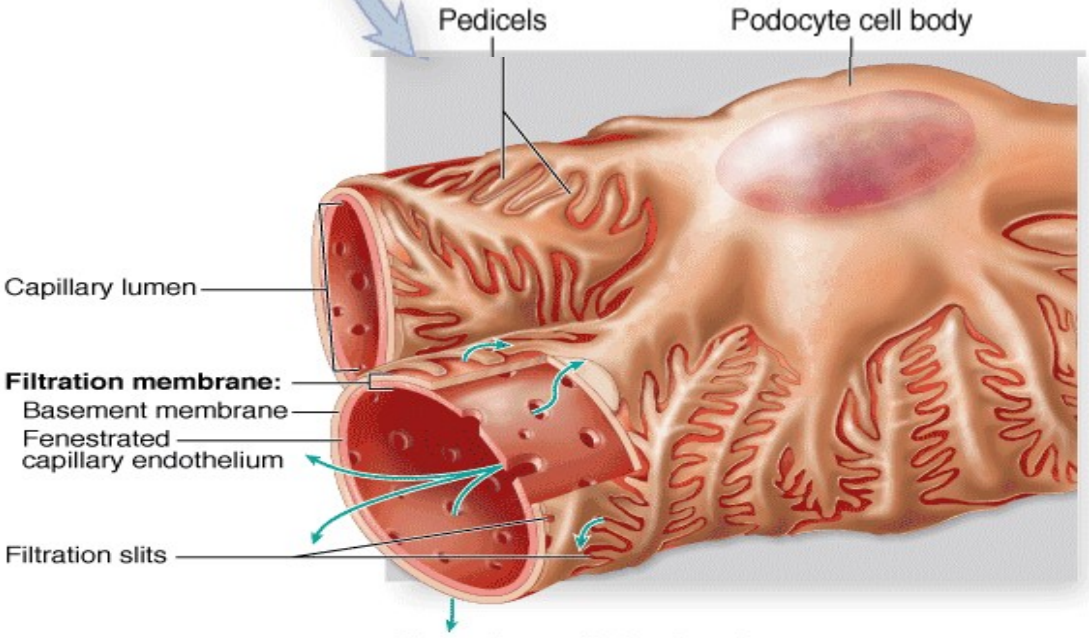


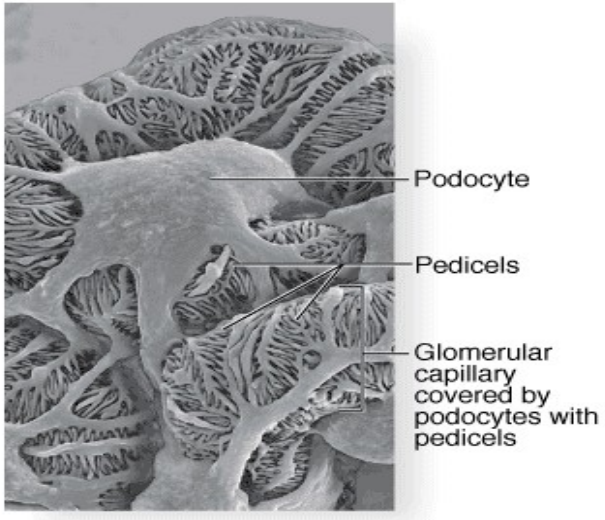


a Renal corpuscle



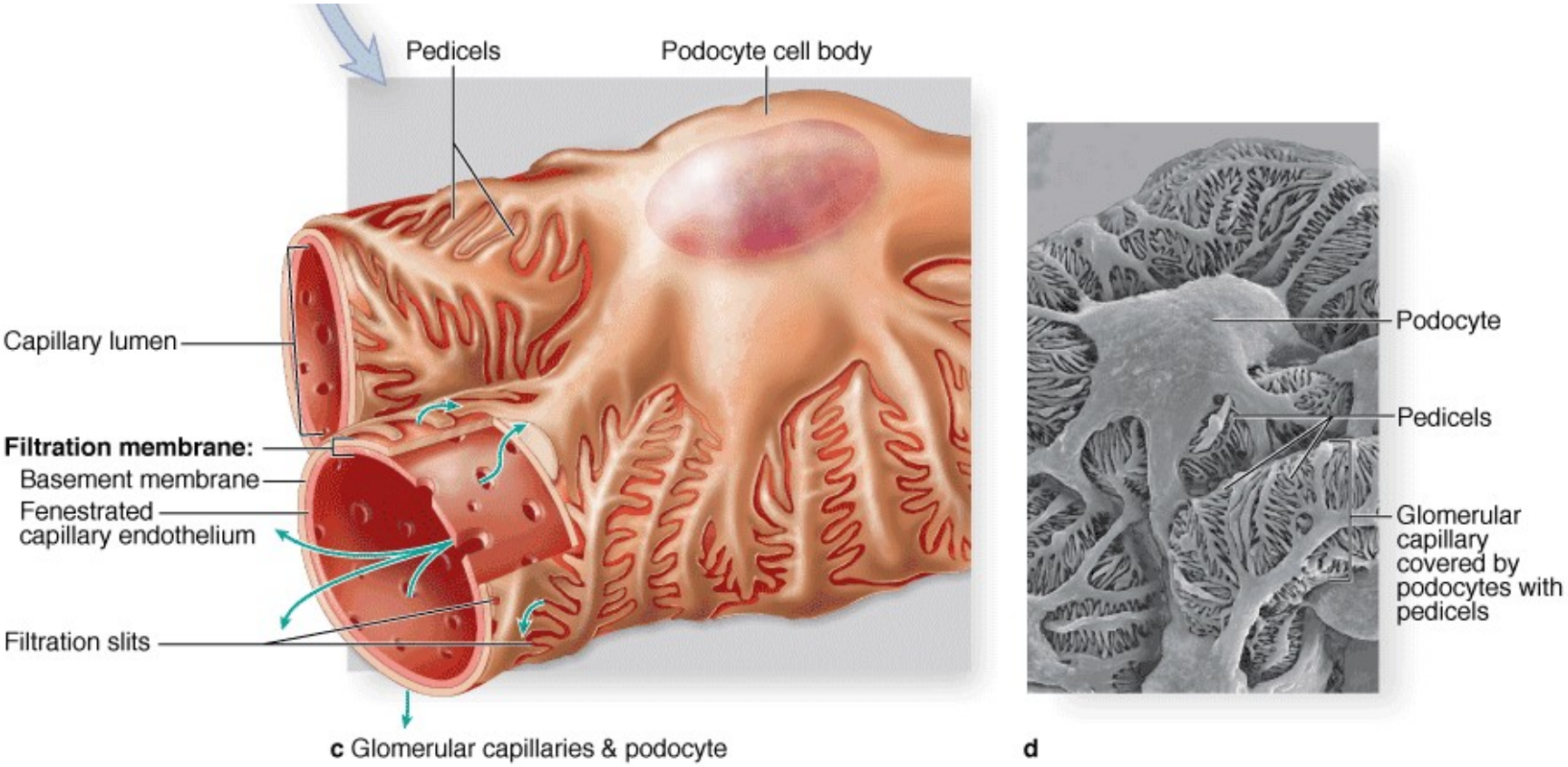
b Histology of renal corpuscle



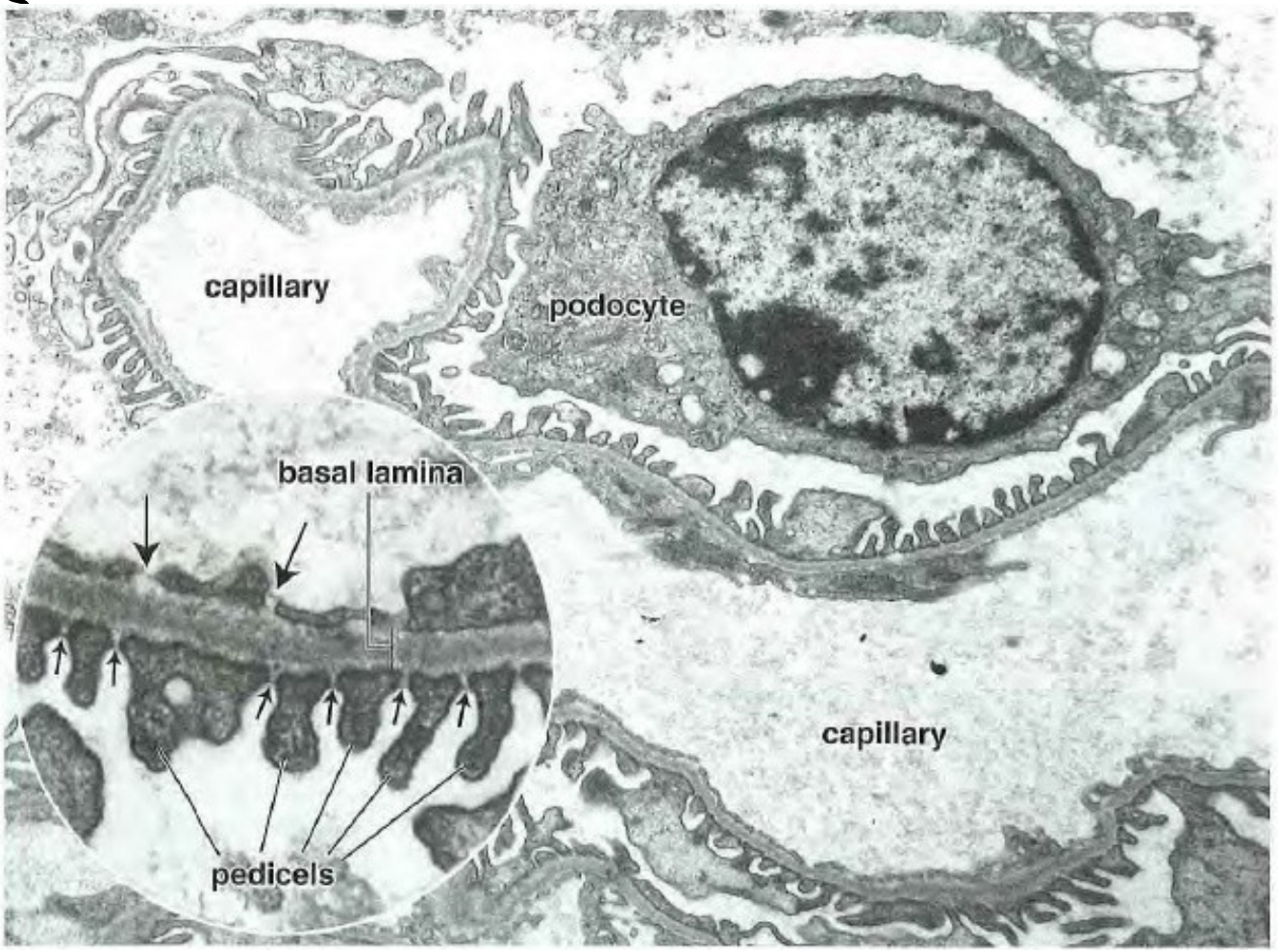


Glomerular basement membrane

- The basal lamina of the visceral layer of Bowman`s capsule that lies between the highly fenestrated endothelial cells of the glomerular capillaries and the covering podocytes is called Glomerular basement membrane.
- It is continues uninterrupted layer.



This membrane is the most substantial part of the filtration barrier separating the blood in the capillaries from the capsular space. It is formed by the fusion of capillary- and podocyte-produced basal laminae and is maintained by the podocytes



It has three layers

- The *lamina rara externa*, adjacent to the podocyte processes. It is particularly rich in polyanions, such as heparan sulfate, that specifically impede the passage of negatively charged molecules.
- The *lamina densa*, the overlapping portion of the two basal laminae, sandwiched between the laminae rarae. It contains type IV collagen, which is organized into a network that acts as a physical filter.

•***Lamina interna.*** The adhesive glycoprotein of this layer help the endothelial cells of glomerular capillaries to keep their attachment to the GBM

Glomerulus

- Glomerular capillaries are uniquely situated between two arterioles—afferent and efferent.
- The space b/w glomerular capillaries is occupied by special type of CT called mesangium.
- The **Mesangium** in the glomerulus form the connective tissue that gives structural support to podocytes and vessels & it consist of **mesangial cells.**
- So in addition to capillary endothelial cells and podocytes, the renal corpuscles also contain **mesangial cells** (Gr. *mesos,* in the midst, + *angeion,* vessel).

- Three cell types make up the glomerulus:
- a) endothelial (red)
- b) mesangial (blue) and
- c) podocyte (yellow).
- Squamous epithelial cells of the Bowman capsule are easily seen(green).
- The macula densa (black) is part of the distal tubule. A collecting duct is in the upper left.



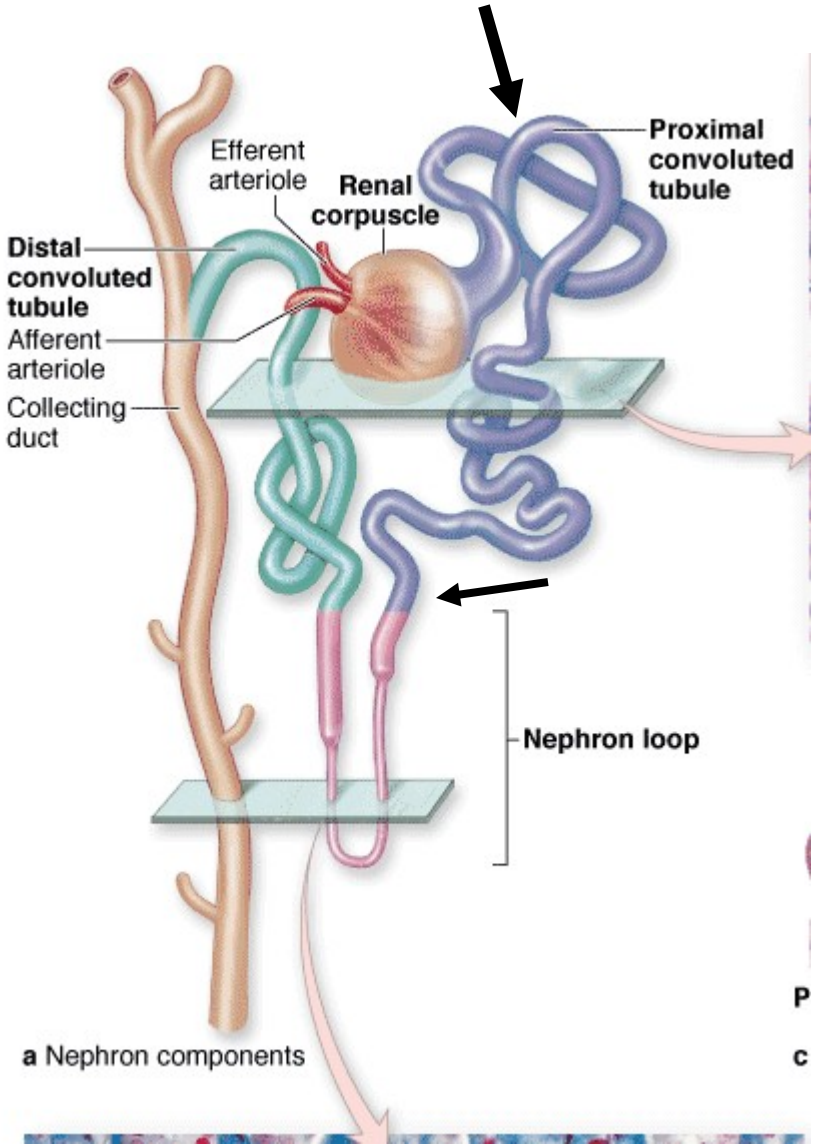
Filtration barrier

- The ***glomerular filtration barrier***, which prevents some large molecules from entering the *capsular space* between the outer and inner epithelial layers of Bowman's capsule.

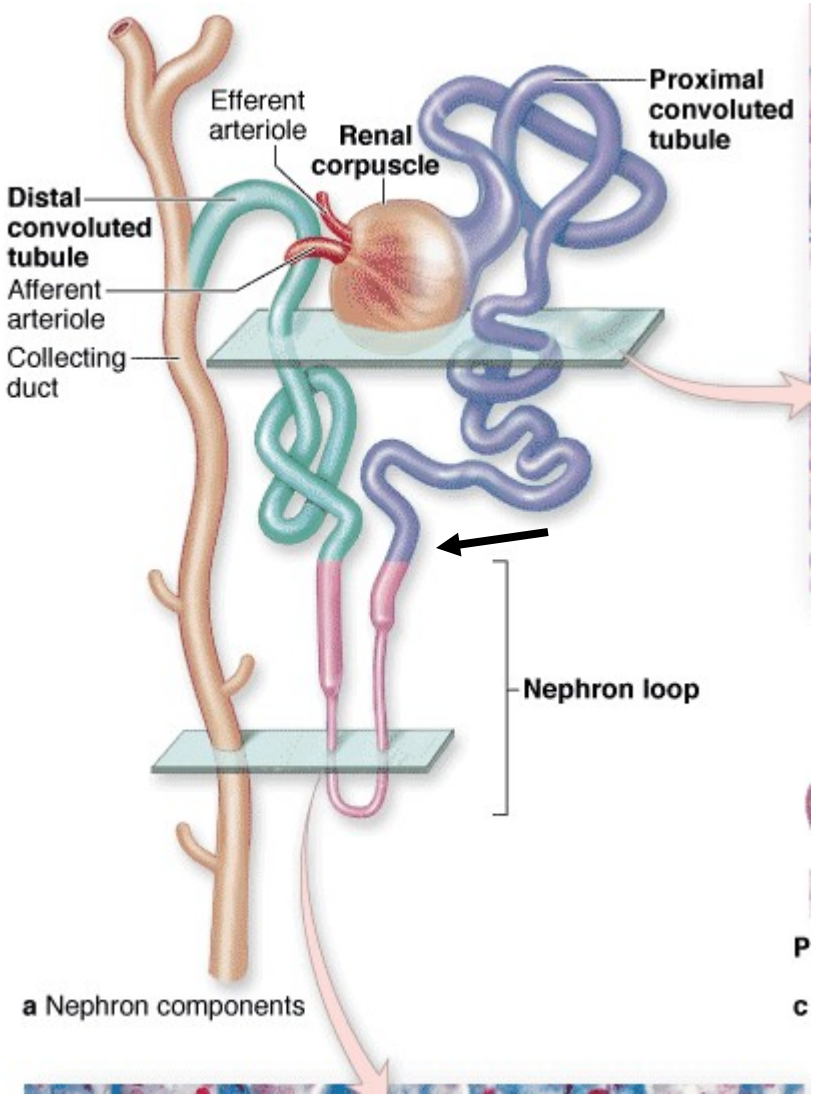
1. The fenestrated capillary endothelium
2. GBM.
3. The filtration slits with slits diaphragm

Proximal tubule

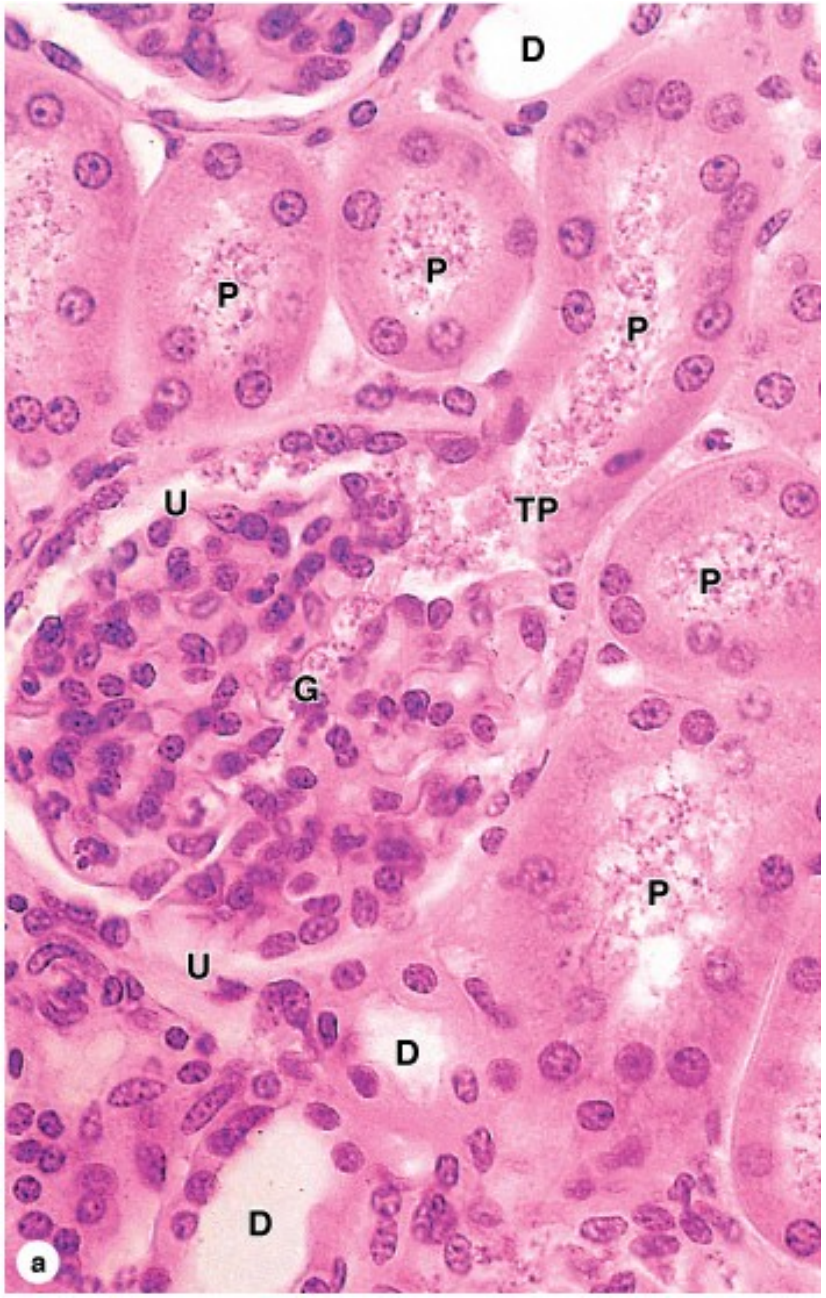
- **The proximal convulated tubule** is the longest section of the nephron (about 14 mm). The convoluted part of the proximal tubules coils close to the glomerulus in the cortex.
- Their walls are formed by **simple cuboidal epithelium.**



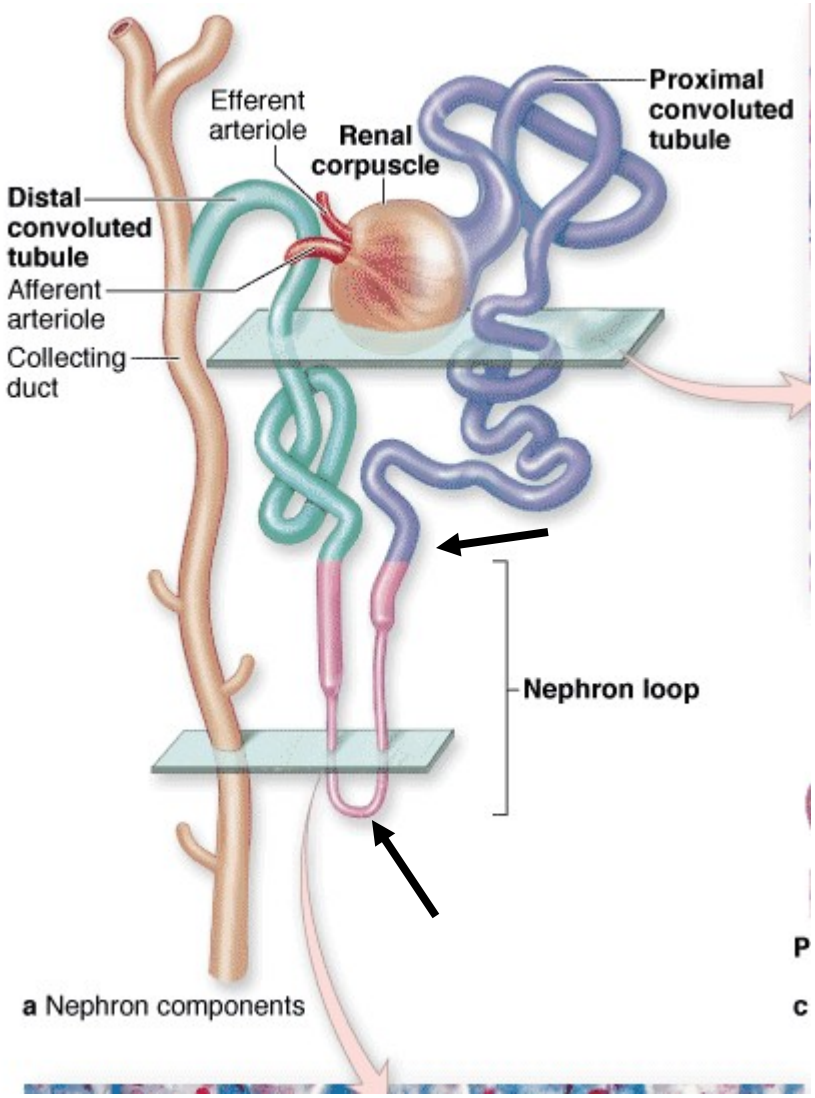
- **<u>Proximal straight tubule.</u>**
- The proximal convoluted tubule continues as a much shorter proximal straight tubule which enters the medulla and becomes the **nephron loop**.
- Their walls are lined by **simple cuboidal epithelium.**



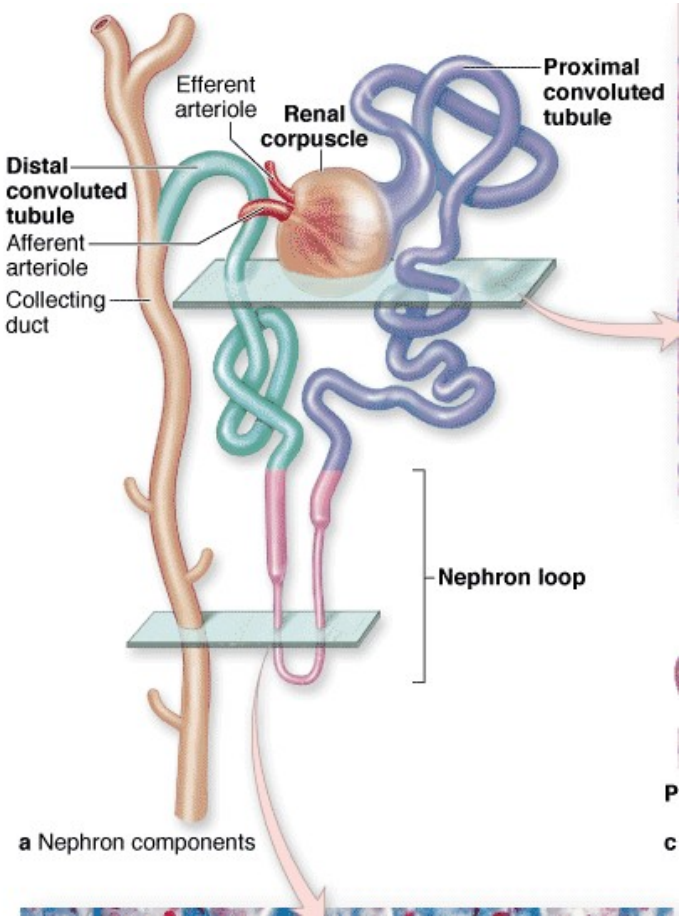
- renal corpuscle's tubular pole (TP)
- urinary space (U)
- simple cuboidal epithelium of a proximal convoluted tubule (P), the long microvilli of the brush border
- distal convoluted tubules (D) appear empty, lacking a brush border



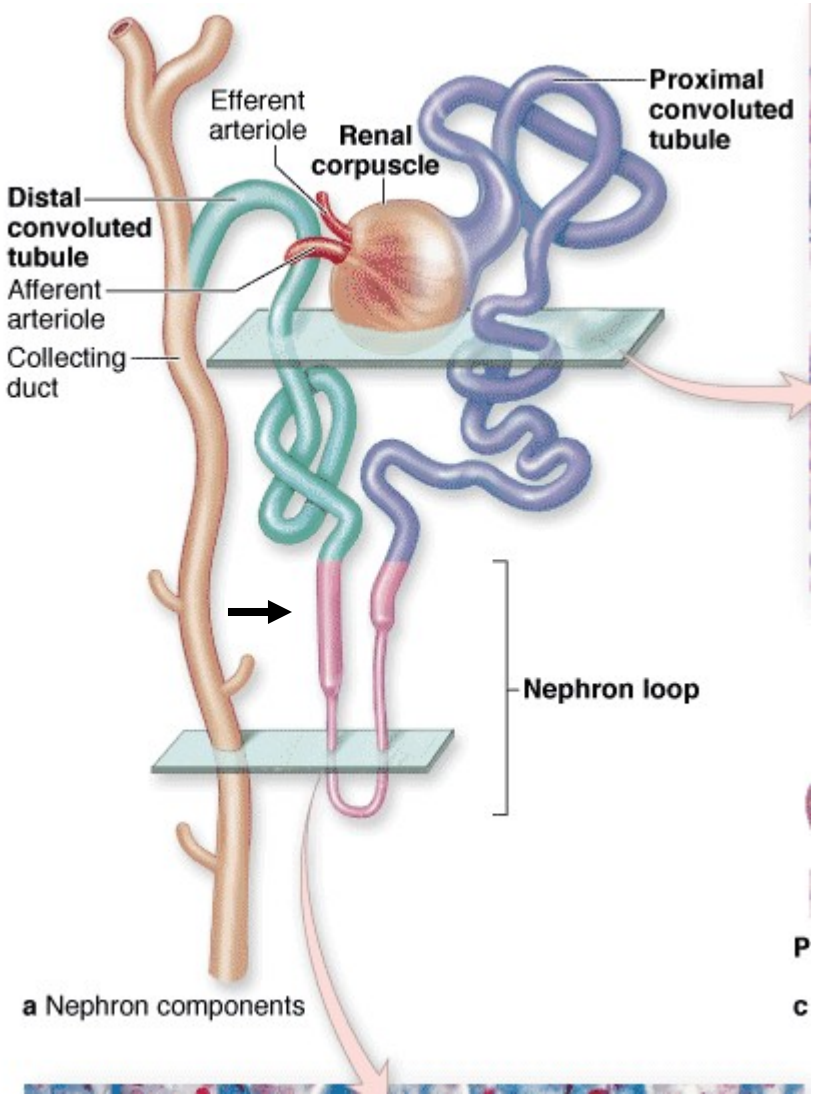
- **Nephron loop or loop of Henle**
- This is a U-shaped structure with a **descending limb** and an **ascending limb.**
- A typical loop of henle consists of the following four parts:
- Thick descending segment(proximal straight tubule)
- Thin desceding segment(intermediate tubule)
- Thin ascending segment (intermediate tubule).
- Thick ascending segment(distal straight tubule)



- PST forms the thick descending segment of limb of loop of Henle. It is lined by **low cuboidal epithelial** cells that has **less well developed brush border** & having **few mitochondria**

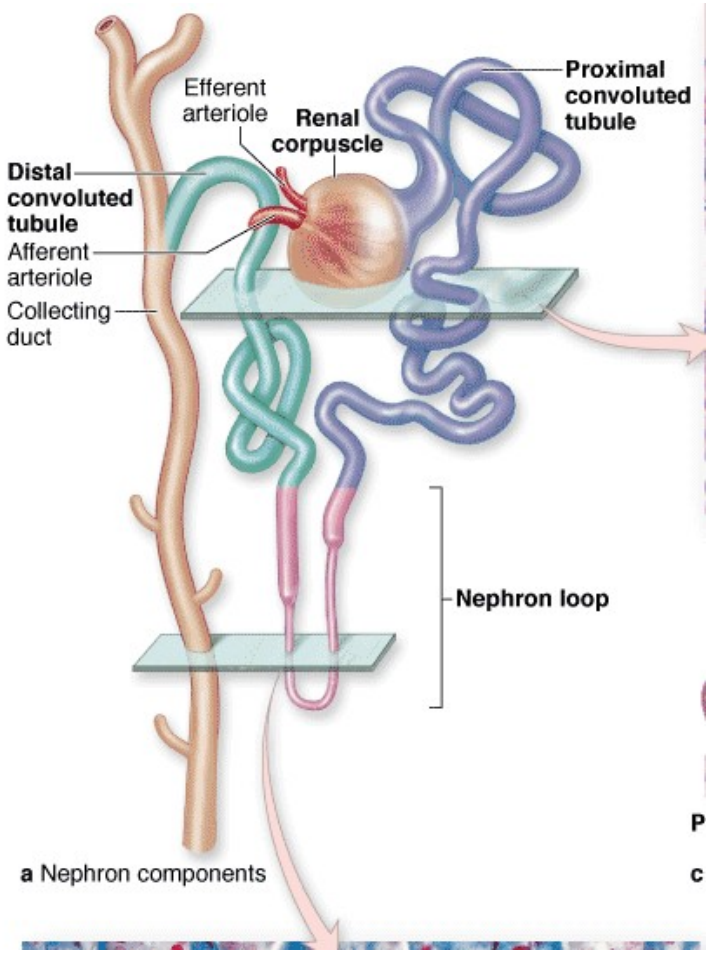


- **The thin segment of Henle's loop** which is linedby **simple squamous spithelium. A few short microvilli** are present.

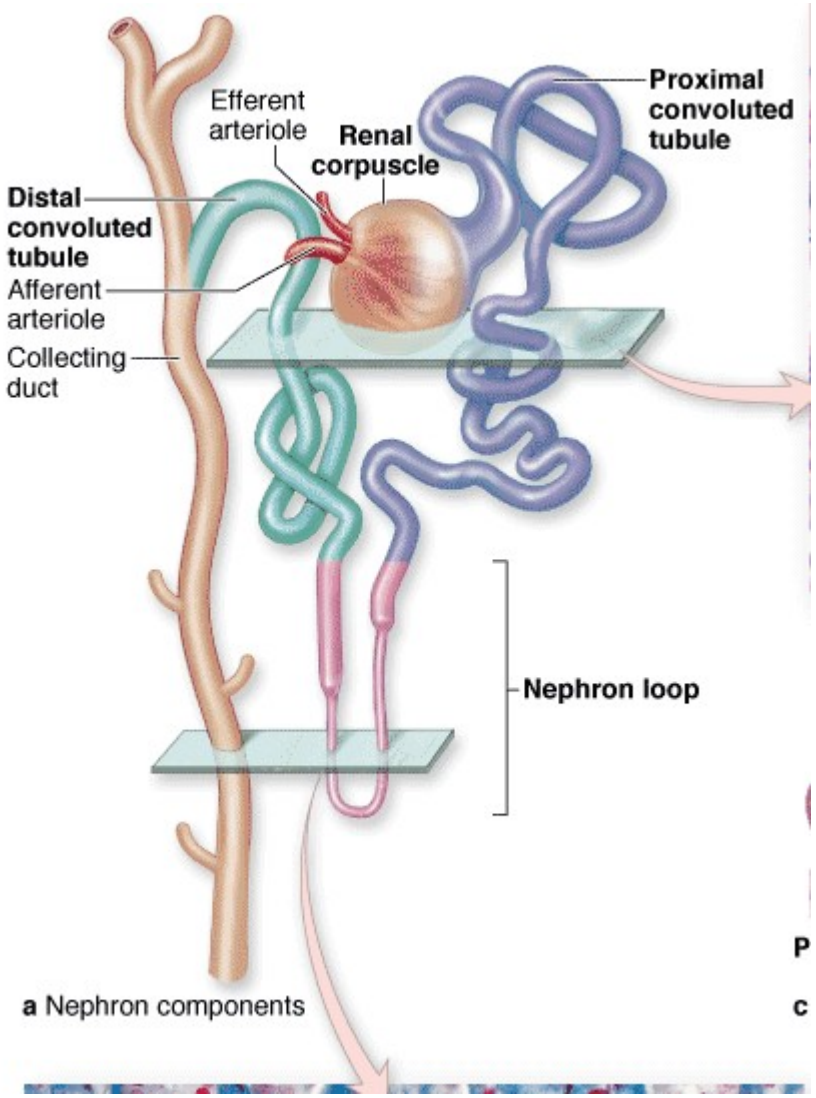


Distal straight tubule

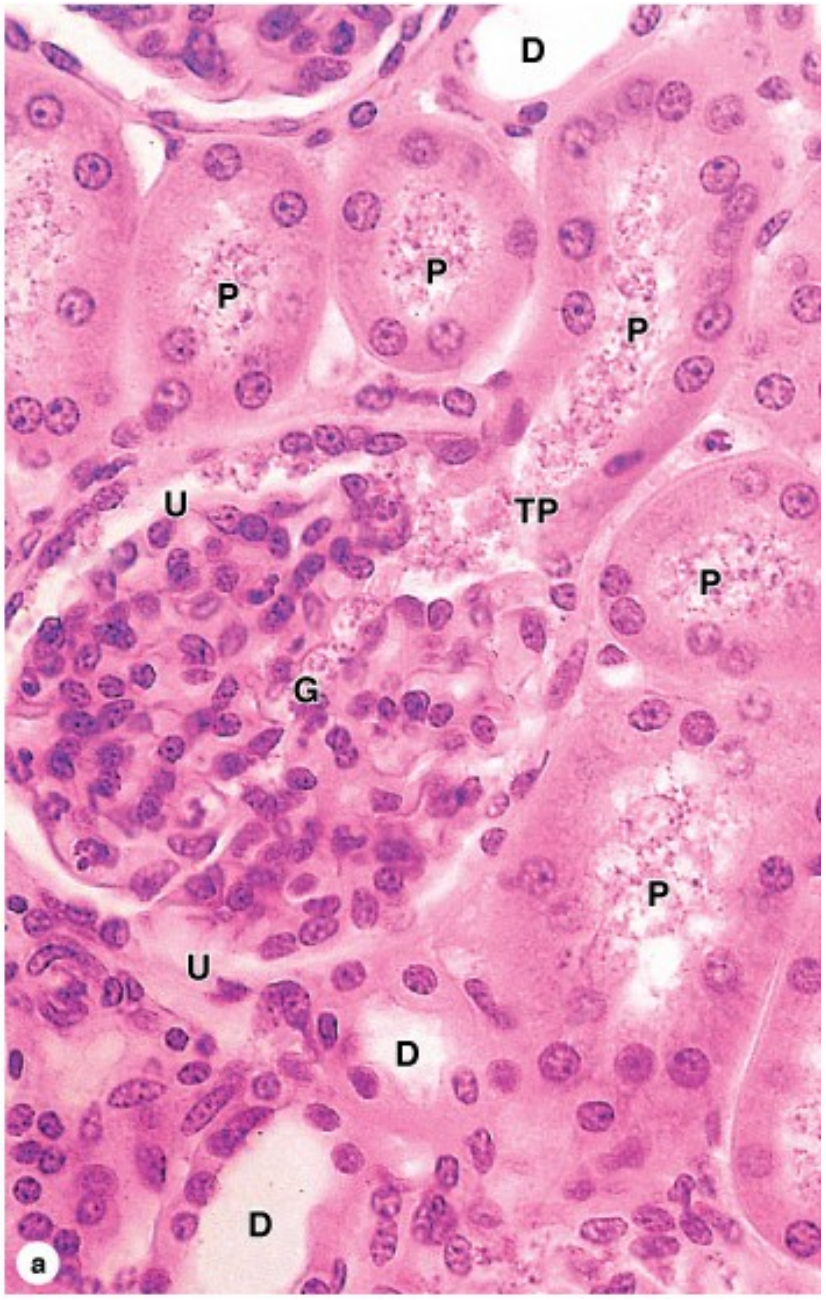
- Lined by cuboidal cells.
- Contain spherical,centrally located nucleus and possess a few club shaped microvilli.

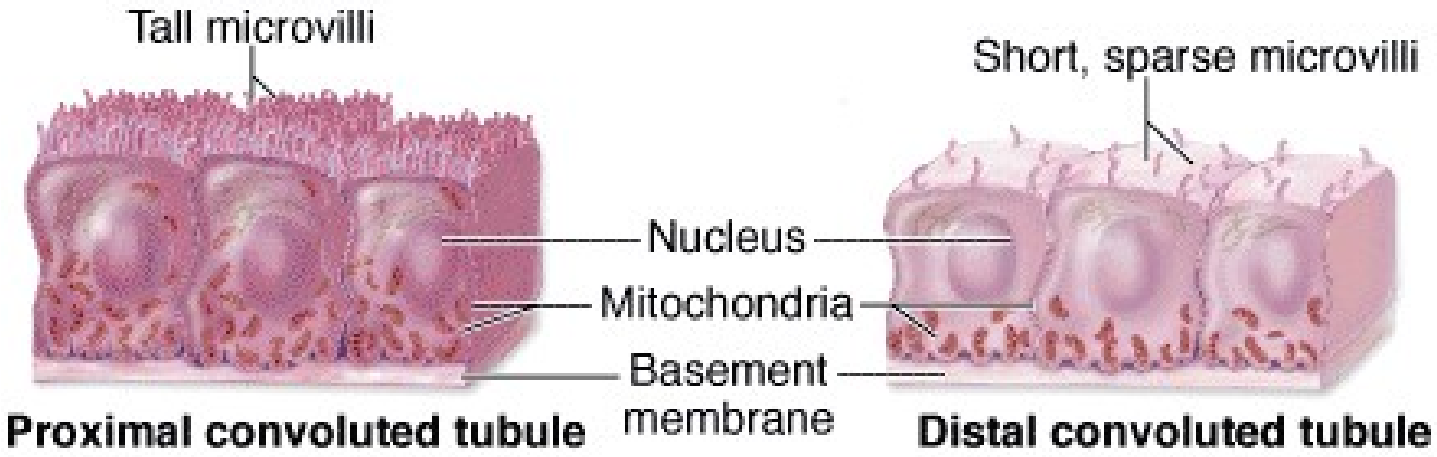


- The **distal convoluted tubule** merges, via *connecting tubules*, with the *collecting ducts*.
- It is lined by **simple cuboidal cells**.
- **Lining cells are thinner then those of PCT but the caliber is larger than that of PCT.**
- **No brush border.**
- It contacts the glomerulus forming the *macula densa*. Thereafter, the distal tubule forms its convoluted portion (about 5 mm long).



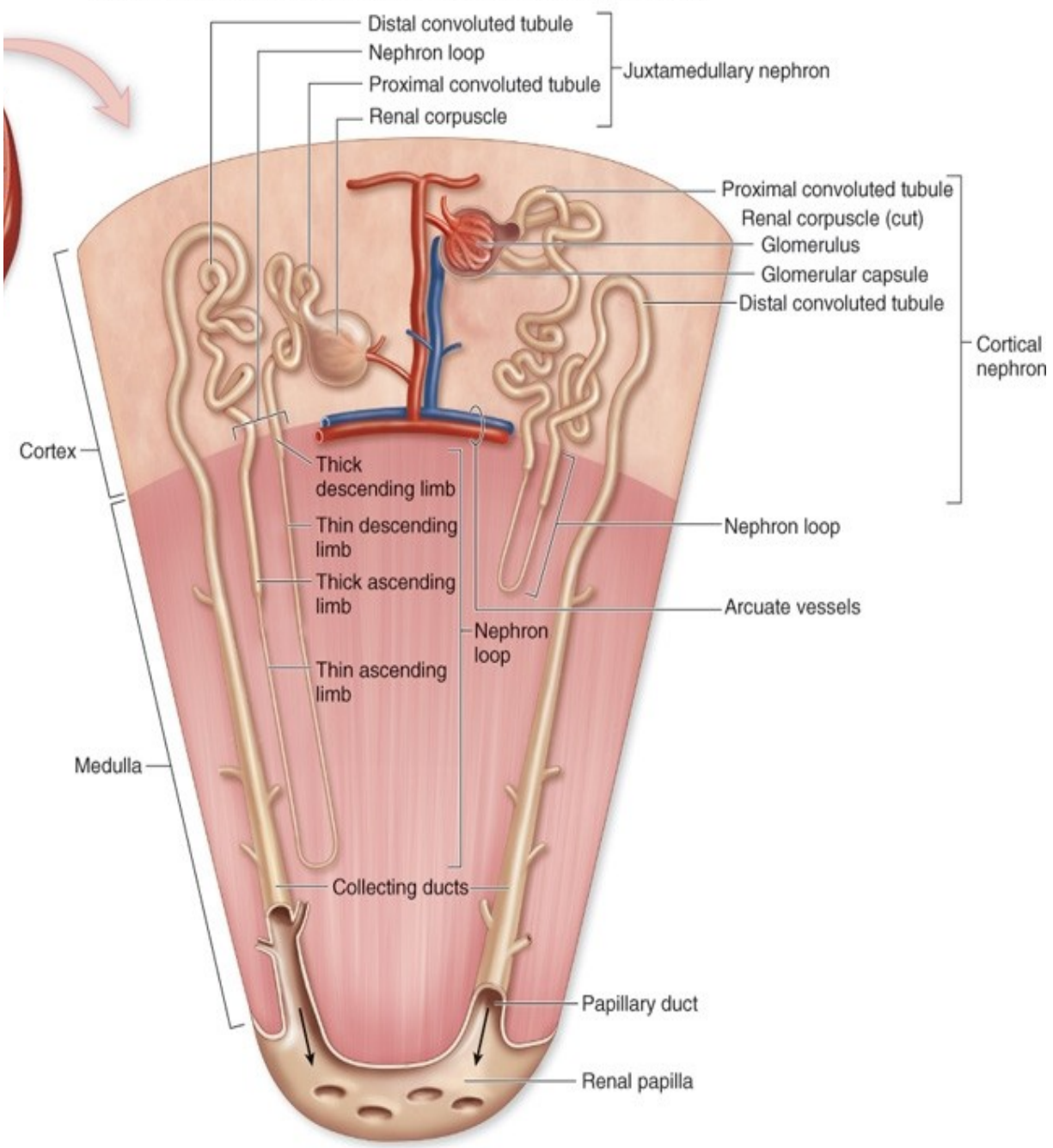
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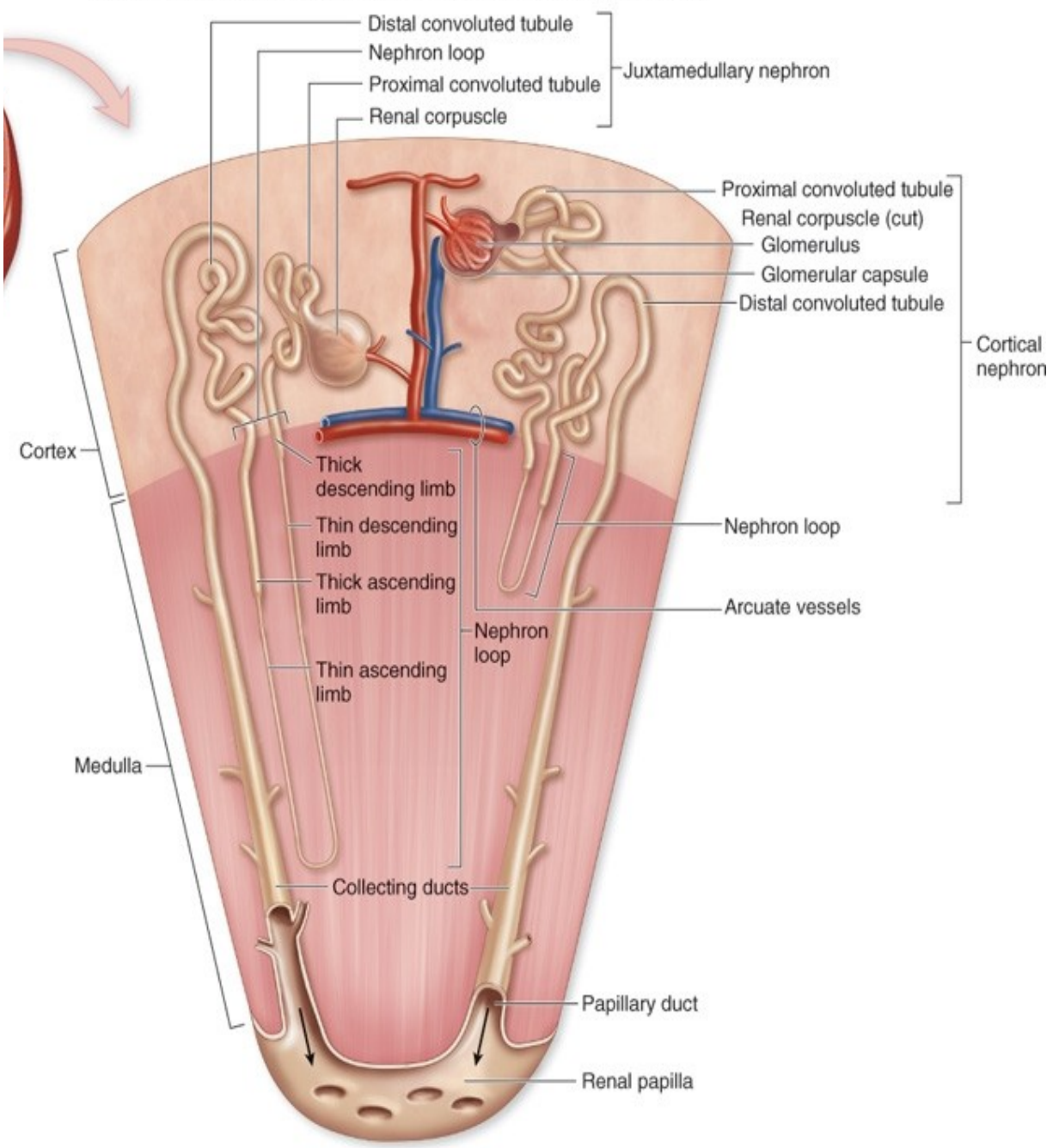

Tall microvilli
Short, sparse microvilli
Nucleus
Mitochondria
Basement membrane
Proximal convoluted tubule
Distal convoluted tubule

Collecting Tubules & Ducts

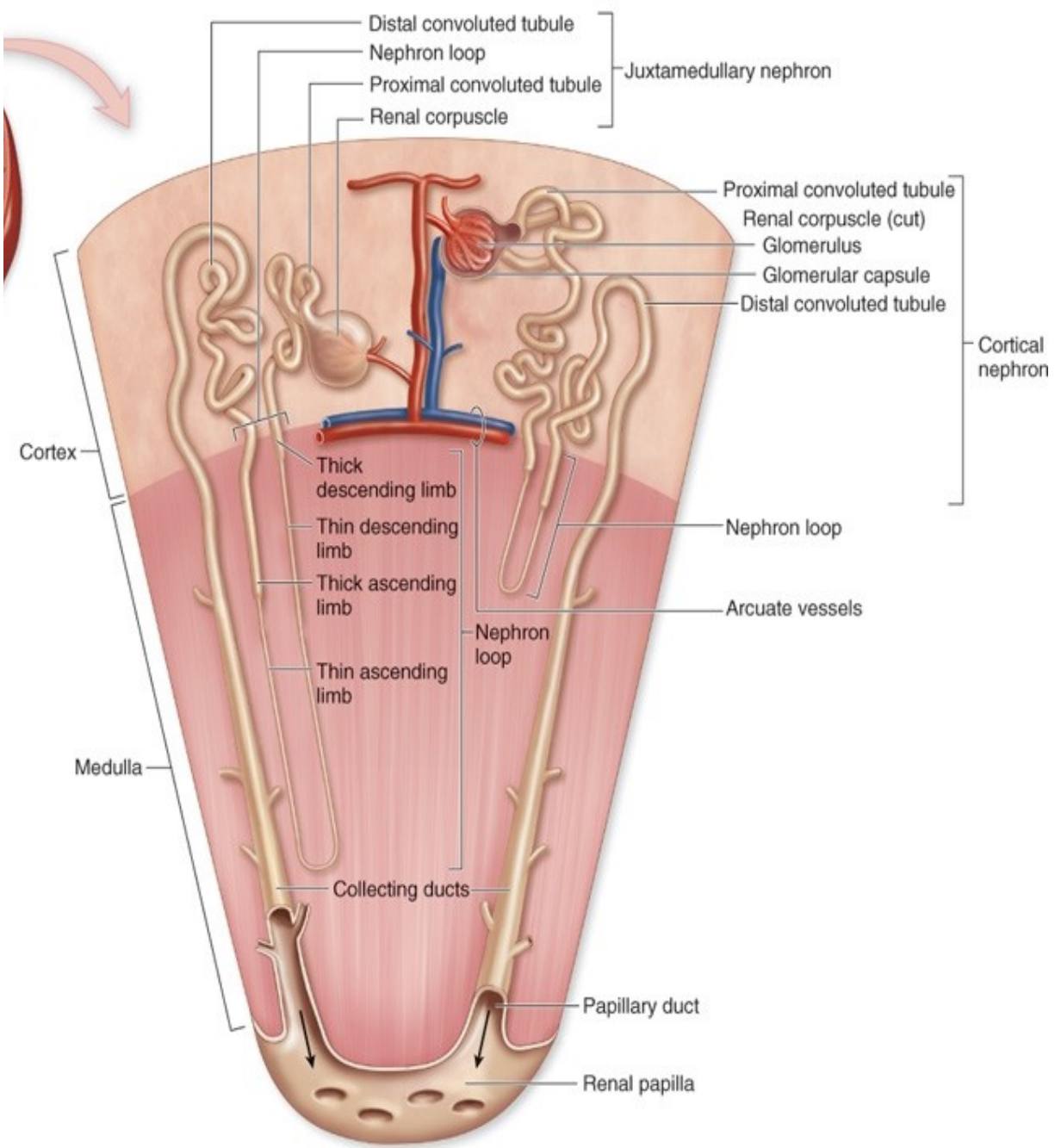
Collecting tubules, the last part of each nephron, which join each other to form larger, straight **collecting ducts**. The collecting tubules are lined with **simple squamous to cuboidal epithelium**

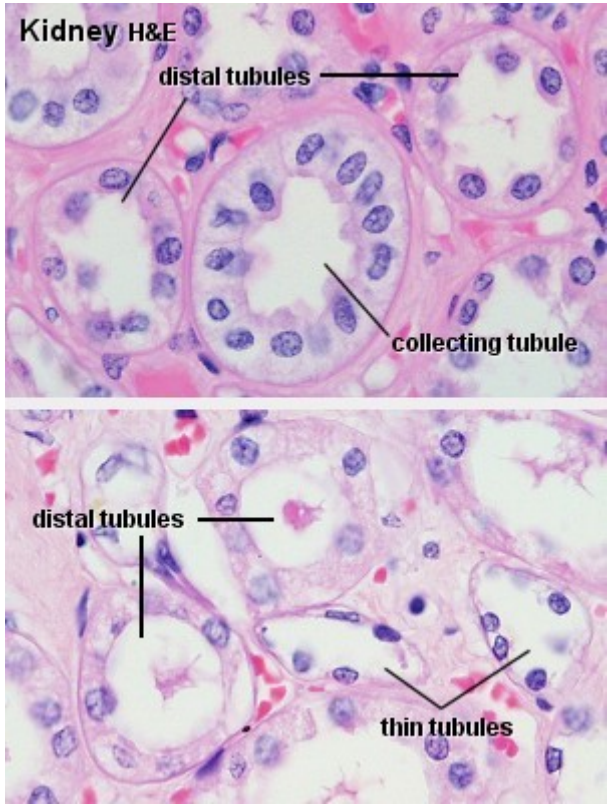


- **Collecting ducts** that run to the tips of the medullary pyramids merge to form **papillary ducts (of Bellini),** which terminate on the tips of the renal papilla and empty into a distended, funnel shaped part (*minor calyx*) of the ureter.



- The collecting ducts lying in renal cortex are called **cortical collecting ducts (simple squamous to cuboidal epithelium**) & the collecting ducts lying in renal medulla are called **medullary collecting ducts (simple columnar**)

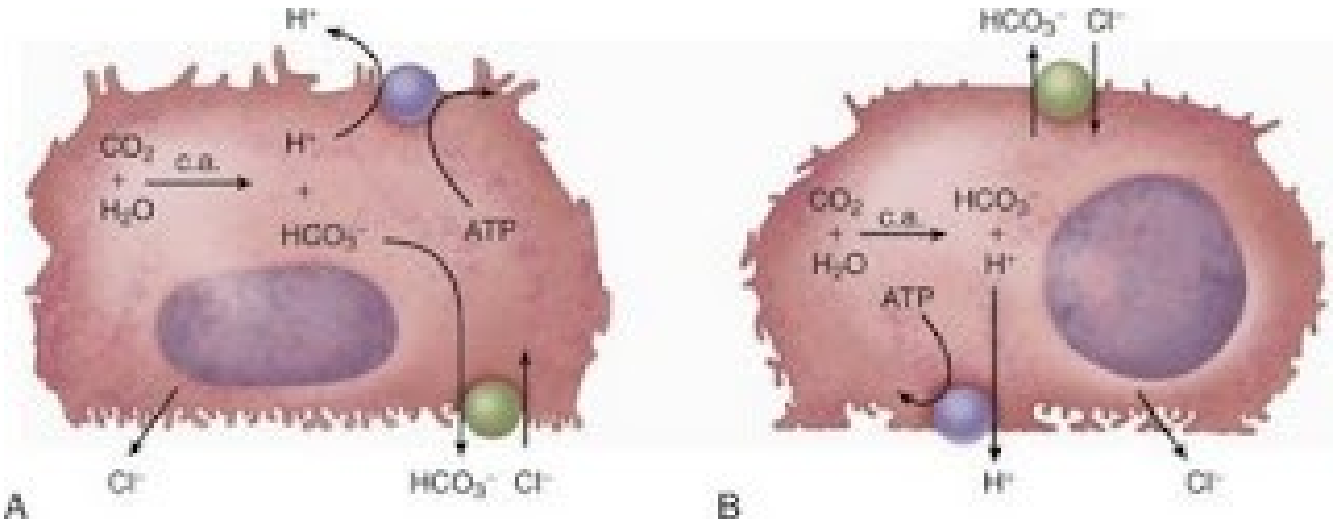



Kidney H&E
distal tubules
collecting tubule
distal tubules
thin tubules

Two distinct types of cells are present in the collecting tubules and collecting ducts:

- ***Light cells,*** also called ***collecting duct or CD cells,*** are the principal cells of the system. They are pale-staining cells with true basal infoldings rather than processes that interdigitate with those of adjacent cells. They possess a single cilium and relatively few short microvilli (Fig. 19.19). They contain small, spherical mitochondria. These cells possess an abundance of ADH-regulated water channels, AQP-2, which are responsible for water permeability of the collecting ducts. In addition, aquaporins AQP-3 and AQP-4 are present within the basolateral membrane of these cells.
- ***Dark cells,*** also called ***intercalated (IC) cells,*** occur in considerably smaller numbers. They have many mitochondria, and their cytoplasm appears denser. Microplicae, cytoplasmic folds, are present on their apical surface, as well as microvilli. The microplicae are readily observed with the SEM but may be mistaken for microvilli with the transmission electron microscope (TEM) (see Fig. 19.19). They do not show basal infoldings but have basally located interdigitations with neighboring cells. Numerous vesicles are present in the apical cytoplasm.

- **alpha & beta intercalated cells of the collecting duct**
- according to the functions the **intercalated cells** classified into two different varieties: alpha-intercalated cells (which secrete acid) and beta-intercalated cells (which secrete base).
- **The alpha-intercalated cells** (called also **A-type intercalated cells** on the left in the figure) are tall, columnar epithelial cells which secrete hydrogen ions into collecting duct lumen.
- **The beta-intercalated cells (B-type intercalated cells** on the right) in contrast are shorter, flatter cells which contain an apical chloride-bicarbonate exchanger (called pendrin) which enables it to reabsorb hydrogen ions & secrete bicarbonate into the lumen of collectind duct.
- The interesting thing is that inducing **metabolic acidosis results in the conversion of beta-intercalated cells to alpha-intercalated cells**--giving the kidney a greater ability to secrete protons and return pH to the normal range.

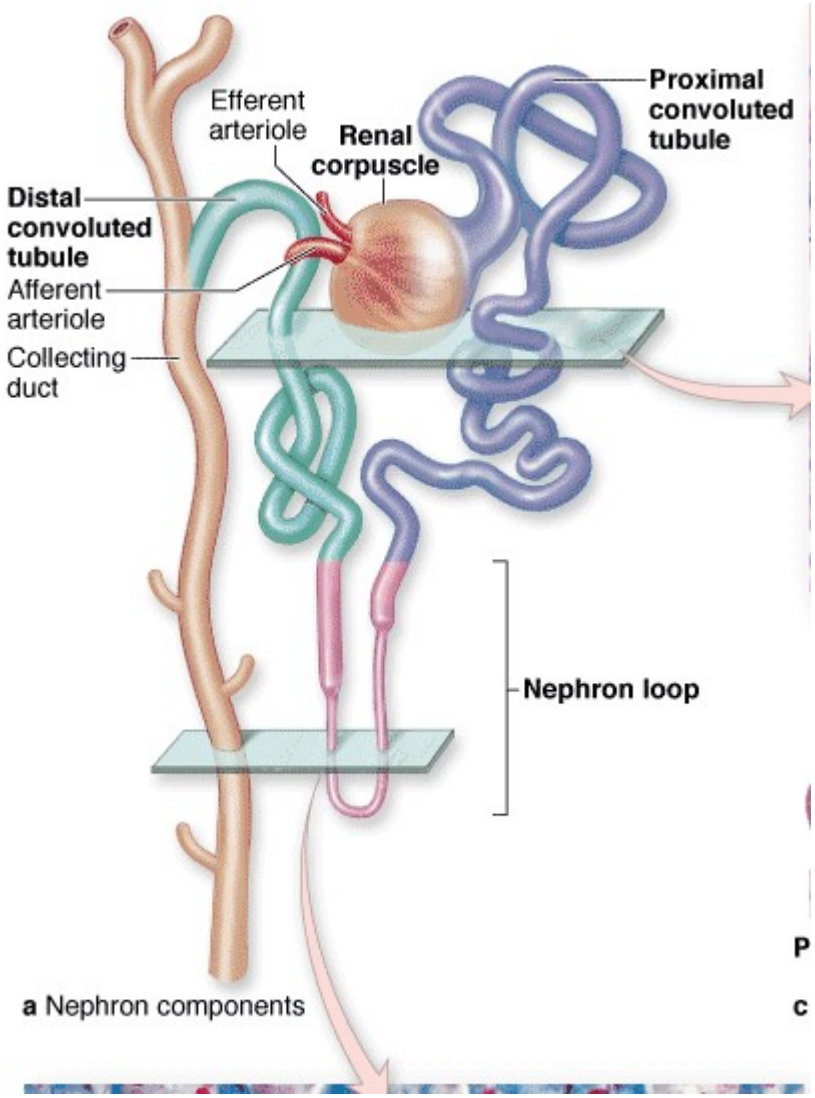


- **The Juxta-glomerular Apparatus** includes

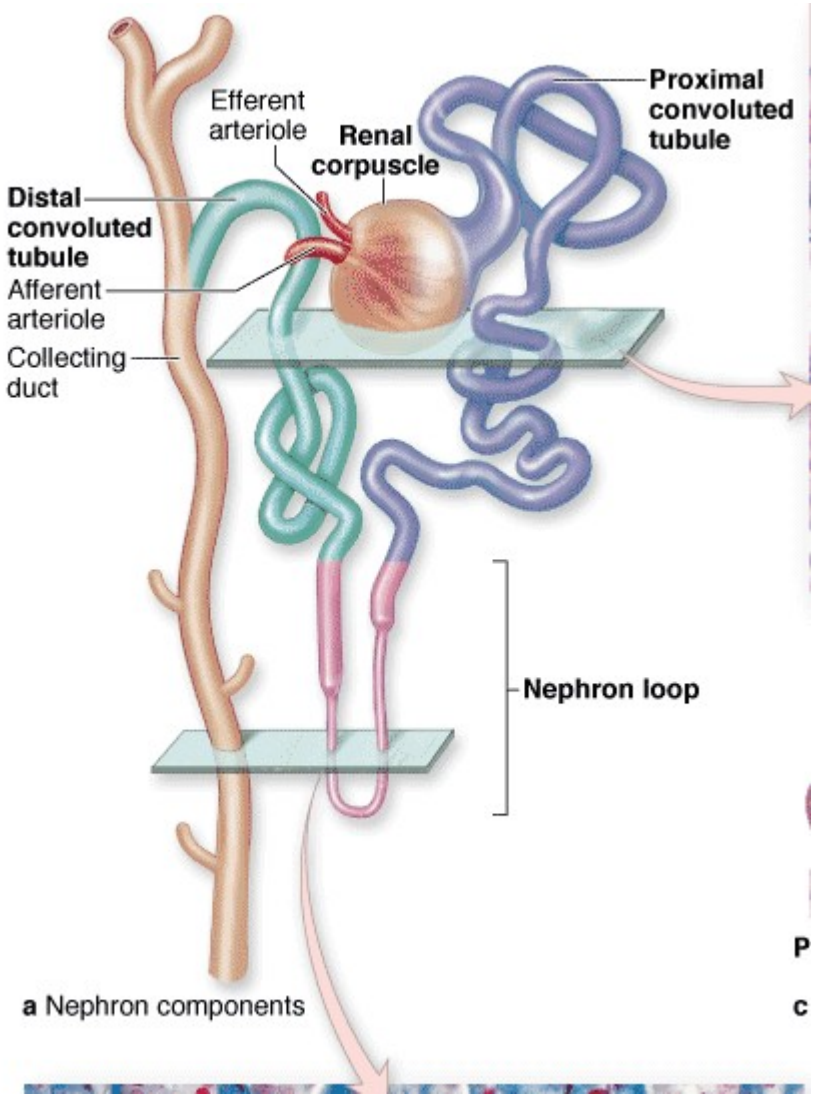
a) juxtaglomerular cells

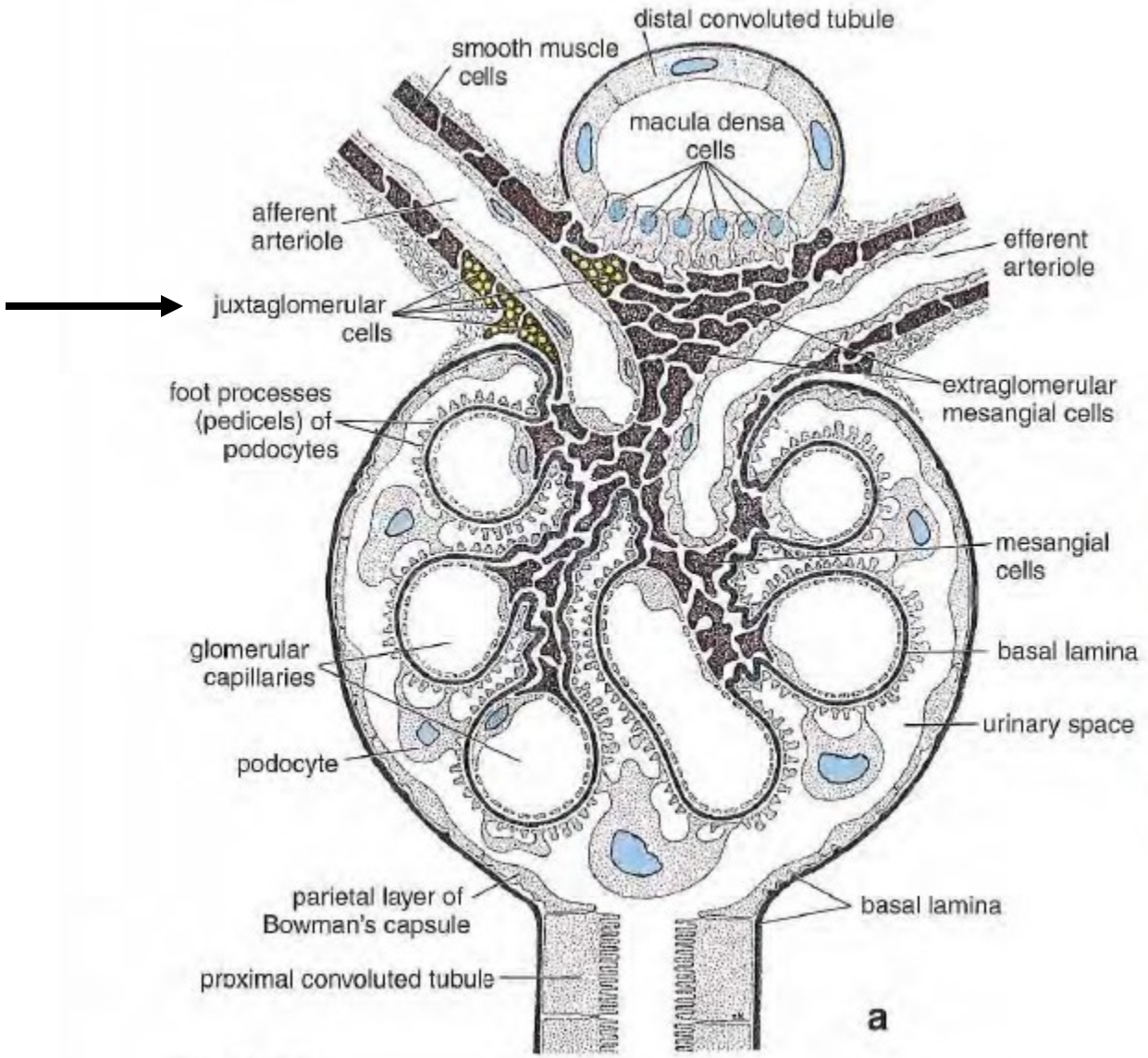
b) macula densa

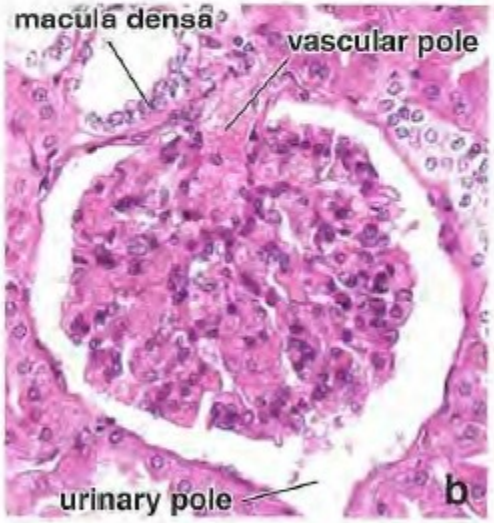
c) extraglomerular mesangial cells



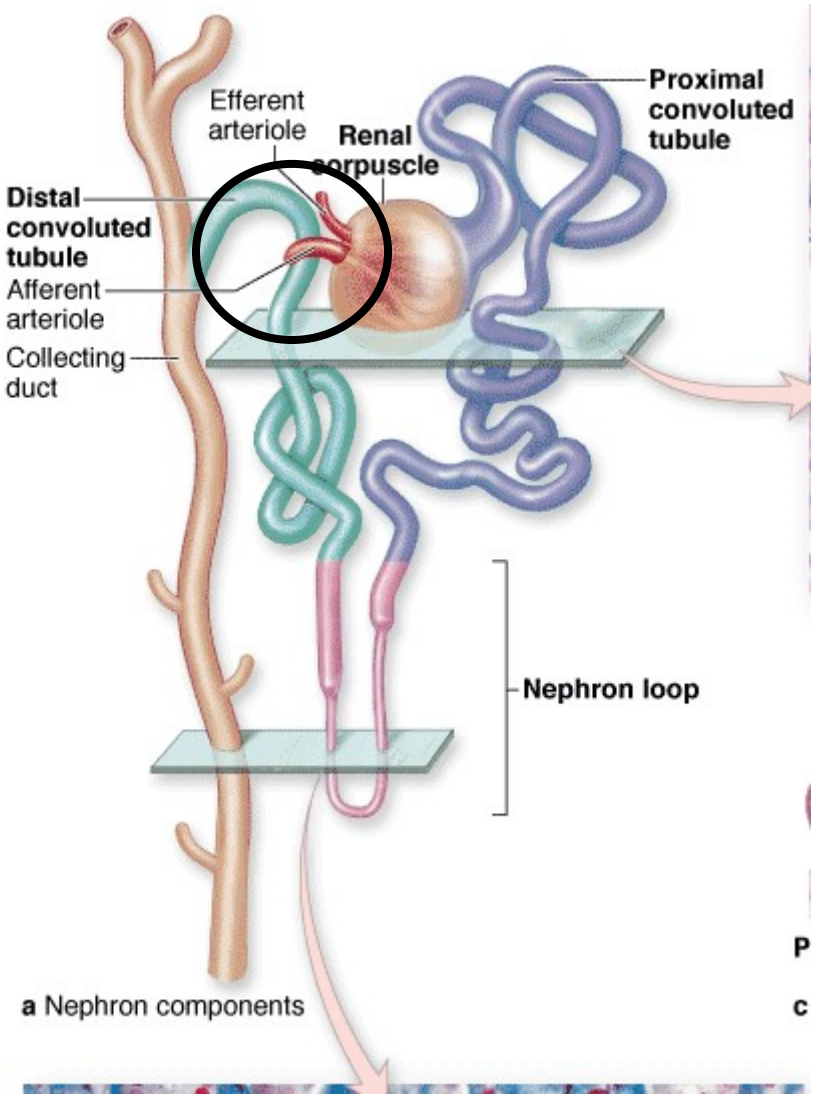
- **JUXTAGLOMERULAR CELLS**
- At the vascular pole adjacent to the DCT (macula densa) the smooth muscle cells in the **tunica media** of the **afferent glomerular arteriole** are replaced by **modified epitheloid like cells with cytoplasmic granules.** These cells are called juxtaglomerular cells which produce and secrete **renin**.

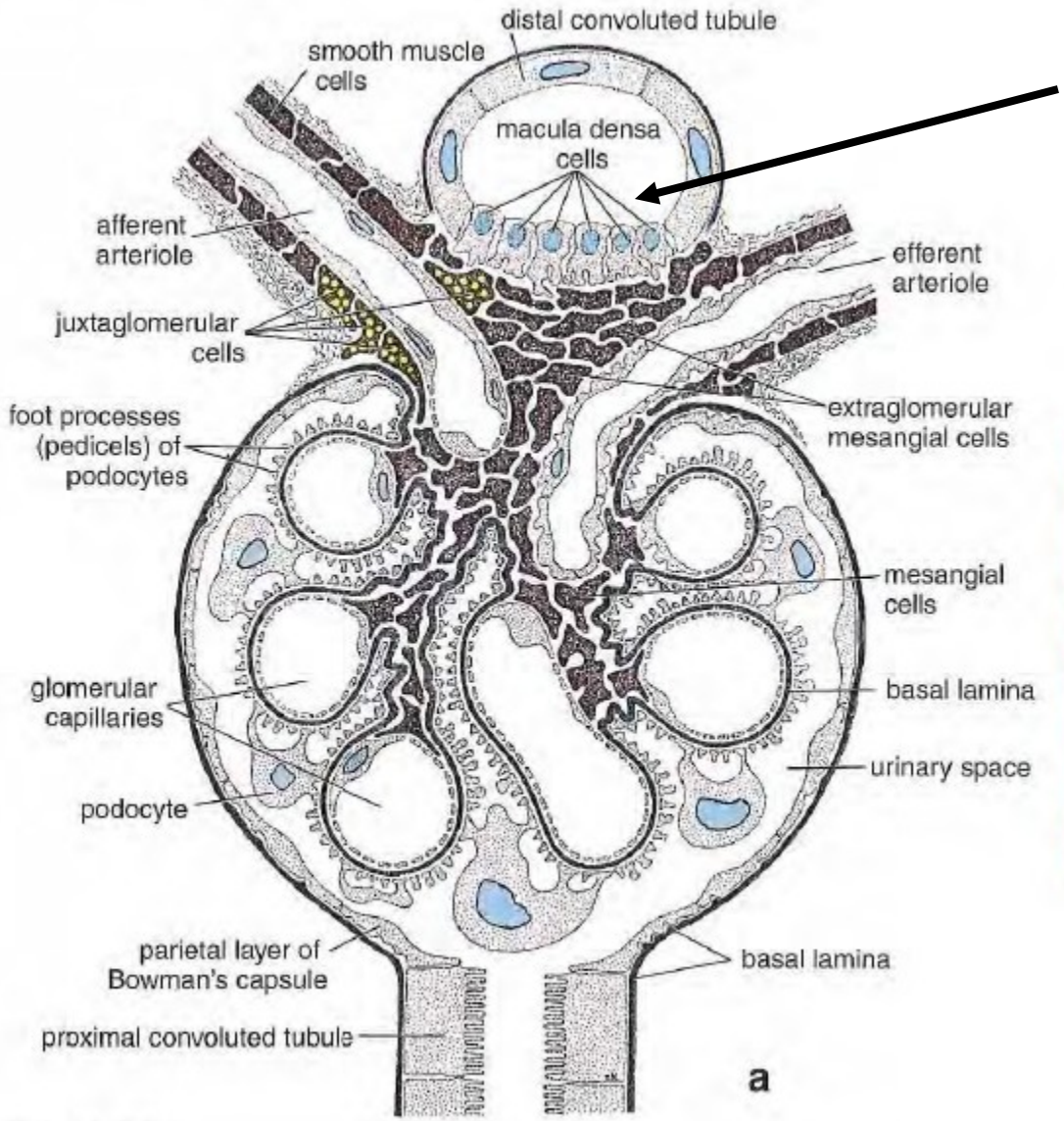


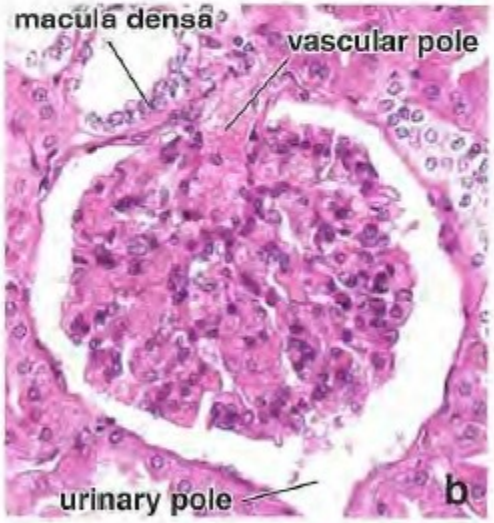

distal convoluted tubule
smooth muscle cells
macula densa cells
afferent arteriole
efferent arteriole
juxtaglomerular cells
extraglomerular mesangial cells
foot processes (pedicels) of podocytes
mesangial cells
glomerular capillaries
basal lamina
urinary space
podocyte
parietal layer of Bowman's capsule
basal lamina
proximal convoluted tubule
a


macula densa
vascular pole
urinary pole
b

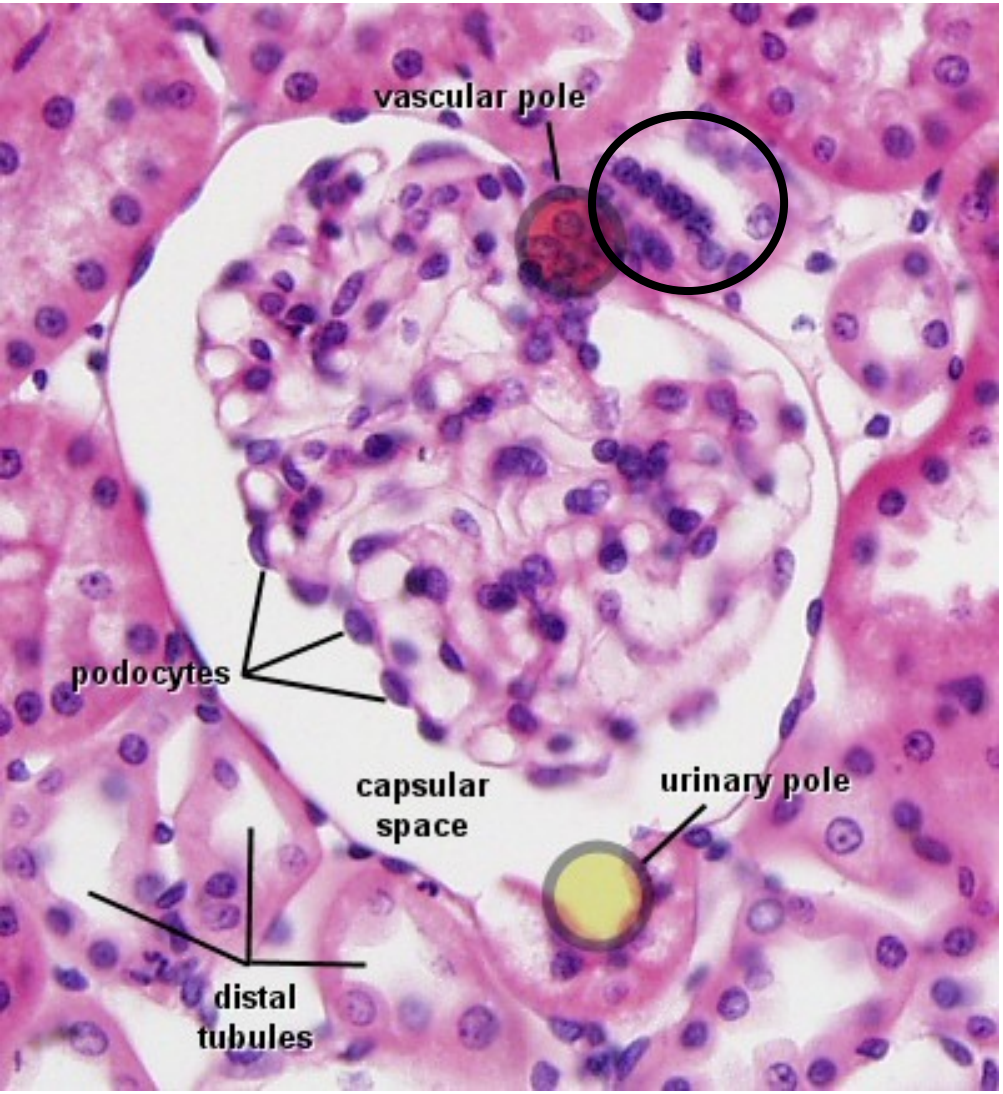
- **MACULA DENSA**
- The DCT contacts the glomerulus. In adjacent DCT, the cells that border the juxtaglomerular cells of afferent arteriole are narrow & more columnar in DCT forming a specialized section of tubular epithelium, the ***macula densa***. This area is darker.
- Cells in the distal tubule are sensitive to the **hormone aldosterone**, which is produced in the zona glomerulosa of the adrenal glands. Aldosterone stimulates the active resorption of sodium ions and the excretion of potassium ions

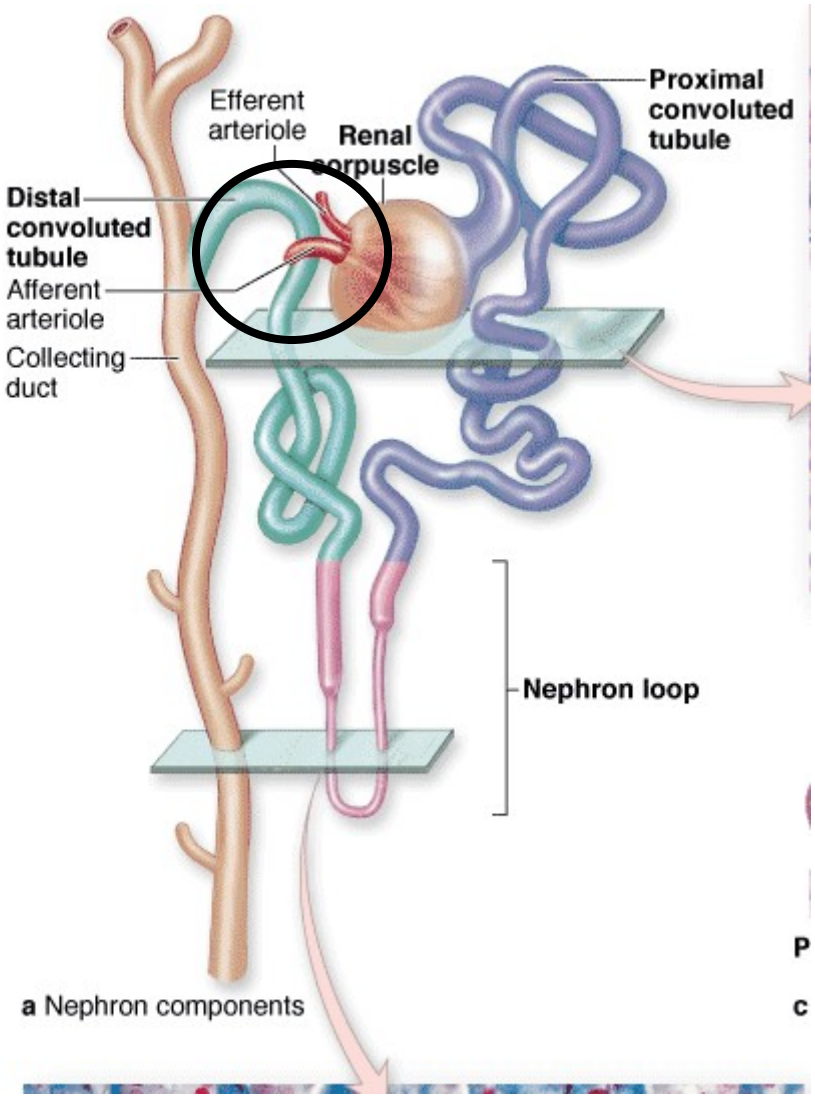



distal convoluted tubule
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efferent arteriole
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foot processes (pedicels) of podocytes
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basal lamina
urinary space
podocyte
parietal layer of Bowman's capsule
basal lamina
proximal convoluted tubule
a


macula densa
vascular pole
urinary pole
b

•Macula Densa

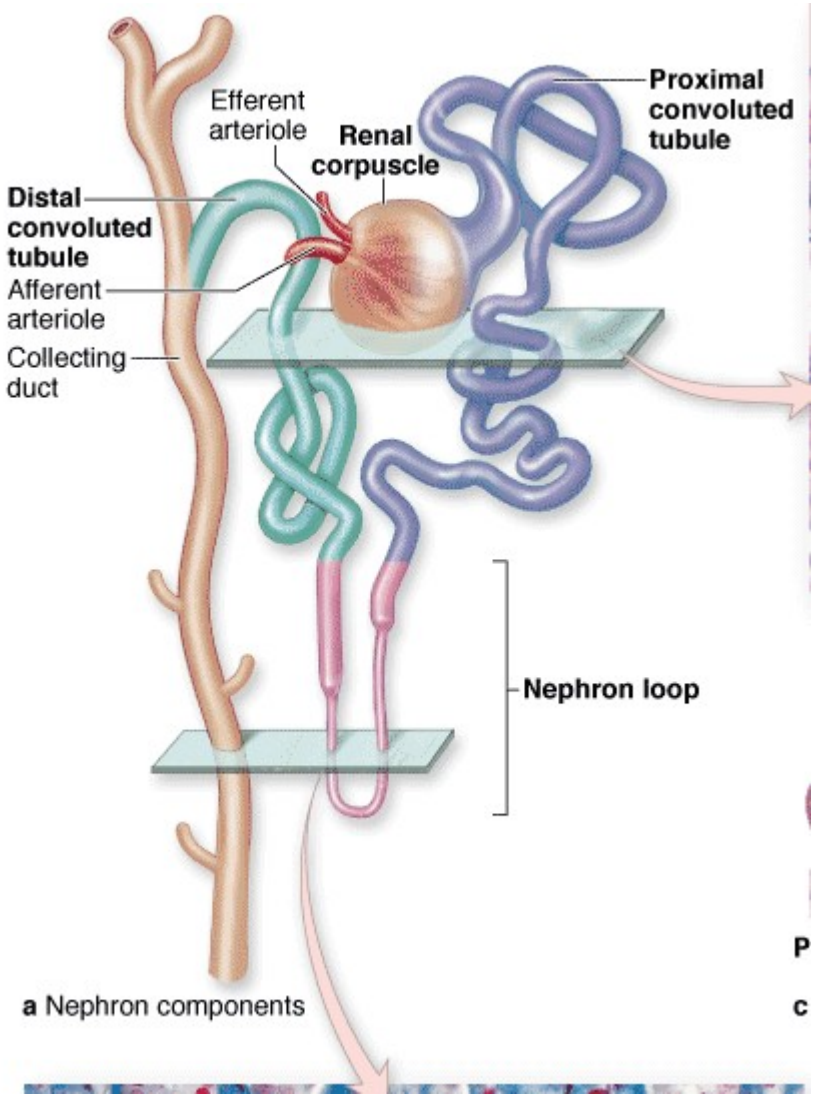


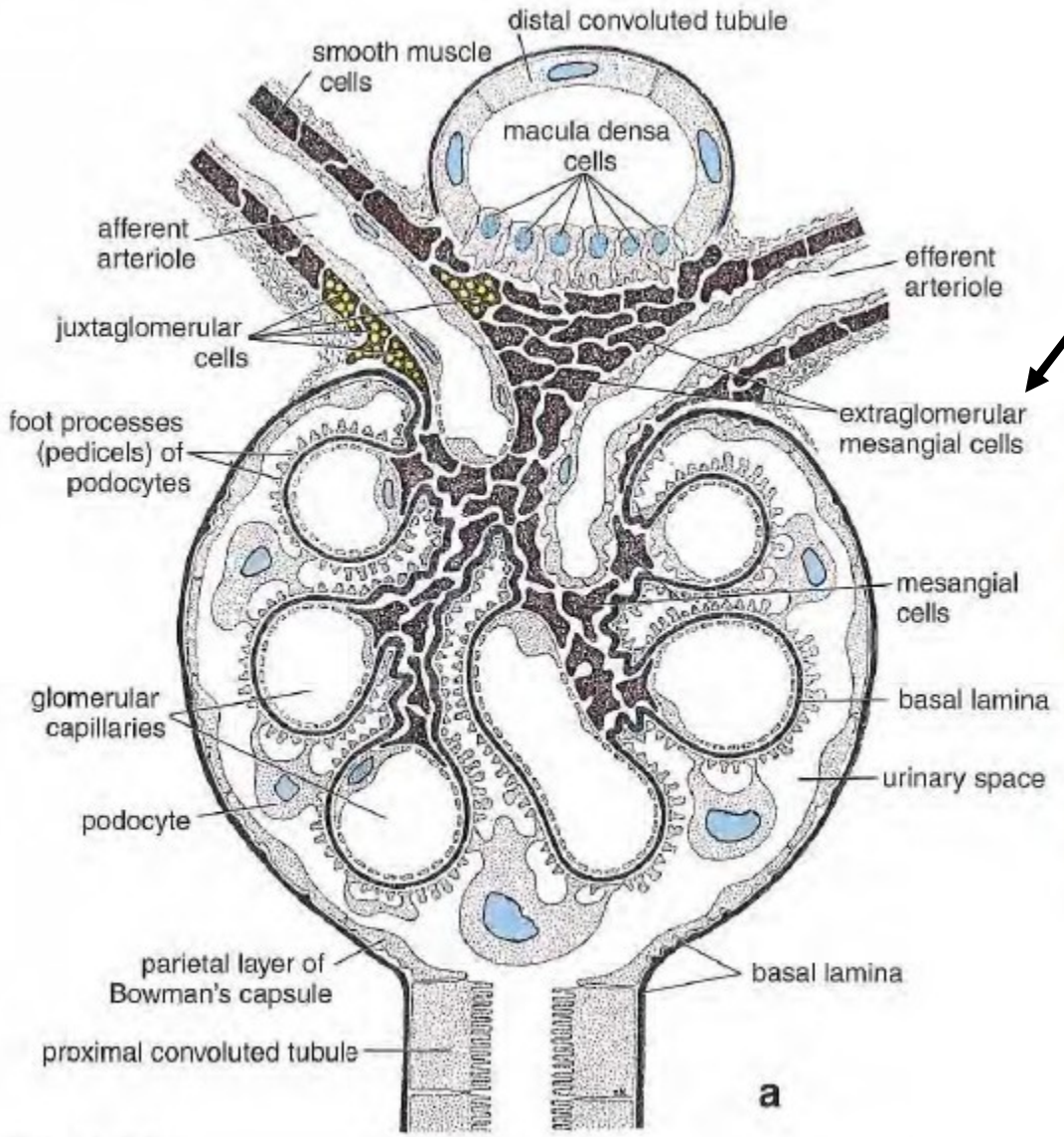


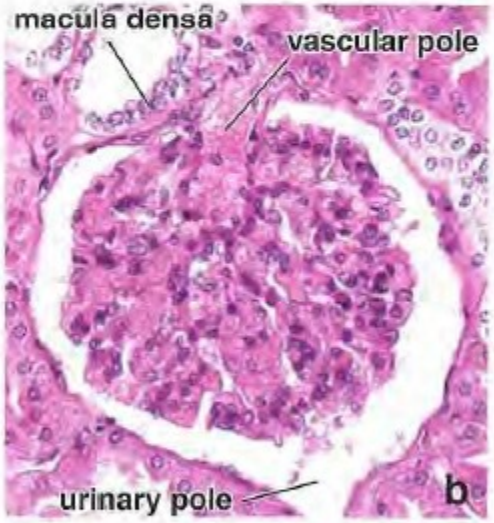
- **Extraglomerular mesangial cells**
- Extraglomerular mesangial cells or (lacis cells/ meshwork) forms a cushion of cells b/w the afferent & efferent arterioles of renal carpuscles. They resemble the mesangilal cells.

Functions

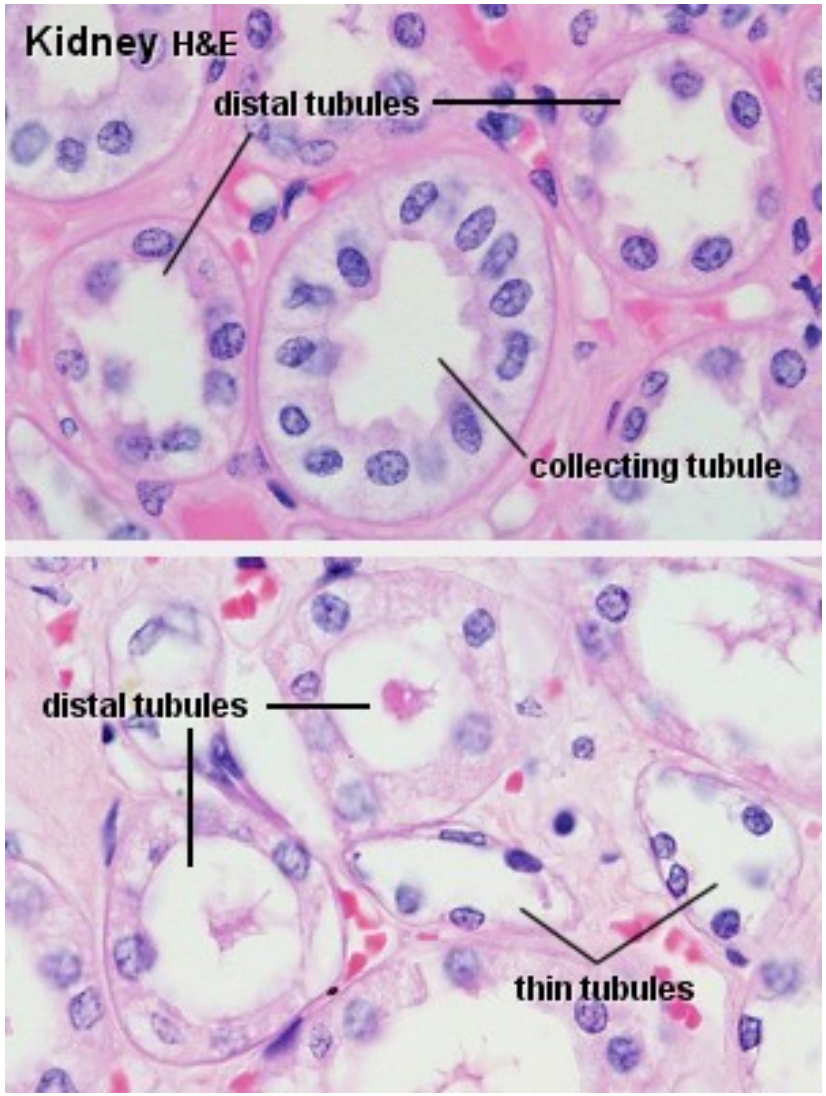
- Supportive functions as these cells inside the glomerulus.
- Lacis cells may also transmit signals from the macula densa into the glomerulus, affecting vasoconstriction there.

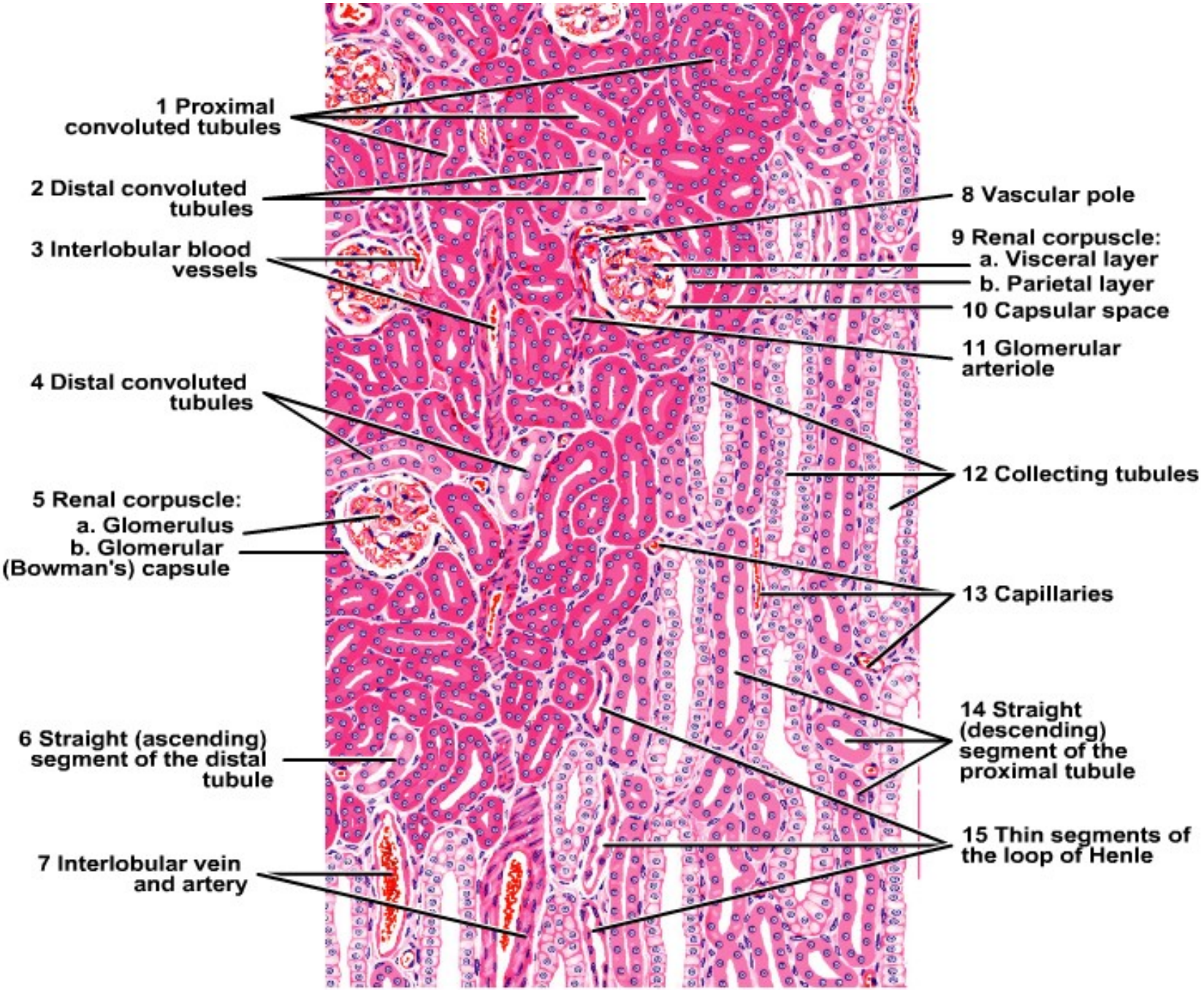



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parietal layer of Bowman's capsule
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proximal convoluted tubule
a


macula densa
vascular pole
urinary pole
b

Collecting Tubules & Ducts
Collecting tubules




1 Proximal convoluted tubules
2 Distal convoluted tubules
3 Interlobular blood vessels
4 Distal convoluted tubules
5 Renal corpuscle:
a. Glomerulus
b. Glomerular (Bowman's) capsule
6 Straight (ascending) segment of the distal tubule
7 Interlobular vein and artery
8 Vascular pole
9 Renal corpuscle:
a. Visceral layer
b. Parietal layer
10 Capsular space
11 Glomerular arteriole
12 Collecting tubules
13 Capillaries
14 Straight (descending) segment of the proximal tubule
15 Thin segments of the loop of Henle